



REFERENCE	ISSUE	DATE
DGT 97996	B	02/05/2016

General Rules

GENERAL RULES FOR USING THE 1553B BUS

SCOPE

Aircraft Data Networks

Direction Générale Technique

<i>Edition</i>	<i>Date</i>	<i>Issue</i>	<i>Author</i>	<i>Visa Rédacteur</i>	<i>Visa Approbateur</i>
<i>Origin</i>	14/12/2004	1	F. COSSON		
<i>Last Edition</i>	02/05/2015	B	G. BLOCH		
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1. PREAMBLE

1.1 Document issues

Date	Index	Author	Updating Purpose
02/14/2004	1	F. COSSON	Initial version of the translation and the adaptation into English of DGT 33750-6
02/05/2016	B	G. BLOCH	Minor updates

1.2 List of updated pages

All the pages of this document are at the latest issue.

Not applicable to this issue

1.3 Summary

This document, in accordance with the “General specifications for using the 1553-B bus” (Document DGT 95088), provides the rules for using the 1553B bus in mission system of Dassault Aviation Aircraft.

The principal specific rules contained in this document concern the following subjects:

- **Data exchange:**

The exchange of data between two items of equipment involves the exchange protocol (transmission of messages on the bus) as well as the associated actions at the beginning and end of the operation (activation of transmission and indication of the arrival of data at the application level).

Two exchange protocols are used: the automatic exchange protocol for the transmission of cyclic messages and the request exchange protocol for the transmission of random messages.

- **Extension of the volume of data to be exchanged:**

Extension of the volume of data to be exchanged is carried out through multiplexing of the transmitting and receiving sub-addresses as well as utilization of the wagon basis mechanism.

- **Bus redundancy:**

All the 1553-B buses are configured to be redundant, i.e. composed of two identical lines. This redundancy is used to cover transmission errors and dialog failures as well as failures of one of the components of the physical structure to increase the availability of the transmission.

- **Synchronization/dating:**

This is obtained by sending a system time and the short cycle start signals.

- **Transfer layer:**

Depending on the nature of the terminals, different services of the transfer layers as defined by the GAM-T-110 (ex GAM-T-103B) can be used. A list of these services is provided in appendix A.

Any deviation from the rules contained in this document must be requested in writing. DASSAULT AVIATION will supply a response to the said request. In addition, any request for deviation must be justified.

2. REFERENCE DOCUMENTS

This document refers to the following documents:

- **DGT 95088 - GENERAL SPECIFICATION FOR USING THE MIL-STD-1553B BUS**
- **MIL-STD-1553B** Aircraft internal time division command/response multiplex data bus. This standard describes the electrical characteristics of the line and its hardware interfaces, the concepts of operation and data exchange as well as the shape of the signal to ensure transmission of digital data over an electrical media used in time-division multiplexing. This standard has been revised (supplements and corrections) (notice 2 dated 08/09/86). The NATO version of the MIL-STD-1553B standard is the STANAG 3838.
- **MIL-HDBK-1553B** Multiplex application handbook. This document provides explanations and recommendations for using the 1553B bus.
- **MIL-STD-1760** This interface standard relates to weapons and associated storage facilities and proposes a certain number of appendices. The appendix dealing with the digital interface clarifies the use of the MIL-STD-1553B bus.
- **ISO 7498-1984 (E)** ISO OSI reference model

3. PRESENTATION OF THE CONTROL SYSTEM

An integrated digital network possesses an architecture made up of several buses and a control system. The networks differ depending on the type of aircraft.

3.1 Operating modes

The operating modes depend on the aircraft.

3.2 Bridges between the buses

The bridges between the buses depend on the aircraft.

3.3 Equipment addressing on the buses

For all aircraft, each remote terminal on the 1553B bus is addressed at the link level using

- A unique **individual address** of between 0 and 30 inclusive.
- A **common address** used for transfers in broadcast mode. This address is 31 in decimal form and is common to all the remote terminals.

General Rules:

- It must be possible to assign an individual address to all remote terminals on the bus for a given system.
- This address will be supplied by the aircraft connector plug, through 7 contacts (5 for the address, 1 for parity and 1 for reference). Any other addresses assigned for equipment connected to the bus must not lead to the terminal or the application being modified or handled.
- Each time it is initialized, a remote terminal must determine and validate its address using parity (uneven parity set up on 7 bits is used).
- A remote terminal must not reply to any messages if its address is invalid.
- It must be possible to address all remote terminals on the bus with this common address 31, as well as with their individual addresses. The bus controller must be able to generate transfers in broadcast mode using address 31, and to be addressed with this number when receiving transfers.
- Although the bus controller does not need to have its own address, it must be given an individual address in cases where it might operate as a remote terminal.

- It must be possible to endow each piece of equipment connected to several 1553-B buses with a separate individual address for each bus. However, each piece of equipment must have the same individual address on a redundant bus from one line to another. (Note: in this document we will refer to “item of equipment”, “piece of equipment” and possibly even “devices”. These are all one and the same).
- Although the bus monitor can monitor all messages on the bus, it must be given an individual address if it becomes necessary to address it specifically during particular protocols in a system.
- On each bus, the ‘0’ address is reserved for dummy equipment. The ‘0’ address is thus not assigned to any real bus-connected equipment. It is used to inhibit items of equipment which cannot recognize their own address, yet which are supposed to recognize any address.
- The terminal for any item of equipment must, for specific maintenance operations, be able to supply the application with the physical address of the connection of this item of equipment to the bus.

3.4 Coupling redundancy

The redundancy of the coupling to the bus must be ensured up to the line transmitter/receiver level. It is desirable that this redundancy occurs up to the encoder/decoder level. It is not obligatory at a higher level (processing logic). Thus, a bus coupler MIL-STD-1553B comprising two transformers, two line transmitters/receivers and possibly two encoders/decoders fulfils the requirements for coupling redundancy.

4. PRESENTATION OF THE TRANSMISSION SYSTEM

4.1 Purpose of the transmission system

The digital data transmission system provides the **communications** services used to exchange data necessary for the execution of processing operations. These operations are divided into basic processing operations. A basic processing operation is defined by all input and output data and by the software block used to carry out operation. The following are essential for the normal functioning of a basic processing operation:

- acquisition of the input data
- activation of the operation
- transmission of the output data

4.2 Identification of necessary services

All processing is divided up among the equipment comprising the Weapon Delivery and Navigation System. In most cases, this equipment does not function autonomously. It is connected to the digital data transmission system which enables it to exchange data.

The first service provided by the digital data transmission system is the **transfer of data**. This concept is described in the paragraph entitled “DATA TRANSFER” on page 11.

The purpose of the WDNS is to carry out operational functions. For a given operational function, several processing operations are involved which are divided amongst the different items of equipment. A mechanism for the synchronization of processing operations must be available for each of the items of equipment connected to the digital data transmission system.

However, for several applications, it is necessary to know the aging of the data exchanged, which means that an information dating mechanism must be available.

The second service ensured by the digital data transmission system is to provide a **synchronization/dating** mechanism. The description of this mechanism appears in the paragraph entitled “SYNCHRONIZATION/ DATING” on page 66.

The third service is the **operational test** of the digital data transmission system. This involves implementing tests that produce repeated transmission errors, equipment inhibition, etc. These tests are described in the paragraph entitled “BUS MONITORING” on page 70.

4.3 Definition of needs

4.3.1 Data transfer

The purpose of the data transfer service is to carry data between application processing operations by transmitting messages.

4.3.1.1 *Techniques for the extension of the data volume to be exchanged*

Given the small number of messages that the 1553-B protocol is capable of transmitting, and the care to be taken during their composition, alternative methods are available to increase the volume of data that an item of equipment can transmit or receive.

The **wagon** technique, already in use on certain aircraft for display dialog, is used once again. Its use has been extended to display dialog and store management. The technique can also be used for other exchanges requiring time-division multiplexing.

The technique involves multiplexing the contents of a message at the application level. With regard to display dialog, each of the contents corresponds to a group of data used to create a page. A single operation, the display pre-synthesis operation, is used to create the group. During the time lapse only one of these contents is transmitted. The same message thus serves to create several different pages.

This technique can be applied generally. It is to be when, for a given processing operation, several different data groups are to be sent at different moments.

Furthermore, certain items of equipment may use the same sub-address to carry data generated by different processing operations. This corresponds to **sub-address** multiplexing.

4.3.1.2 *Implementation of extension techniques selected*

Each of these two techniques requires additional control data to be included in the data phase.

With regard to the system, a simplification of the message is sought for these two techniques so that no difference between them can be perceived during monitoring of the messages sent on the bus.

With regard to the equipment, the wagon technique is implemented at the application level. However sub-address multiplexing is implemented at the terminal level.

4.3.2 Synchronization/dating

The mechanisms to be implemented require a reference, either a time reference (system time value) or a reference using signals.

In all cases, the processing operations carried out to guarantee the consistency of the data required by the synchronization/dating mechanisms are carried out centrally, i.e. at the level of the equipment that contains the time reference.

4.3.3 Test of correct operation

A standardization of messages transmitted to check the correct operation of the network is required.

In order to avoid increasing the complexity of the terminals of certain items of equipment, sub-address multiplexing will not be used for the messages thus defined.

5. DATA TRANSFER

5.1 Data entities

5.1.1 Overview

5.1.1.1 Equipment modeling

5.1.1.1.1 System objectives

The purpose of the equipment is to perform basic processing operations which when linked carry out basic functions such as navigation piloting and Man Machine Interface or operational functions such as different firing as well as operations relating to system maintenance. These various basic operations are known as **application processing operations** (or simply application processing).

An item of equipment thus consumes and produces data. When the source or the destination of this data involves different items of equipment, **communications functions** designed to perform data exchange are provided for each item of equipment. These functions provide communications services.

The objectives to be taken into account for the transmission system are:

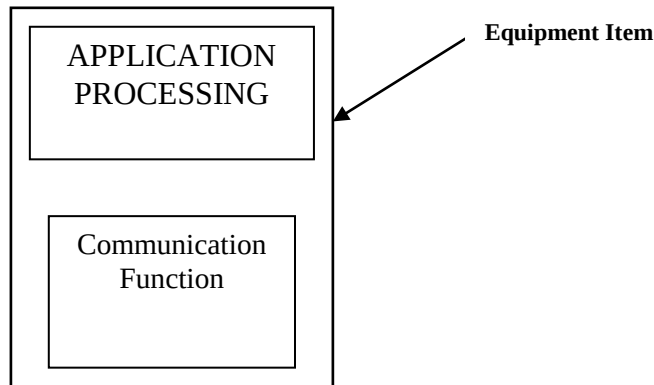
- Ease in validating and integrating items of equipment
- Upgrade flexibility, i.e. is it possible to modify the communication functions irrespective of the application processing operations of the equipment.

These considerations entail, firstly, a definition of the communications functions common to all the items of equipment and, secondly, a proposal for an equipment model with which the different functions will be associated.

5.1.1.1.2 Description of the model

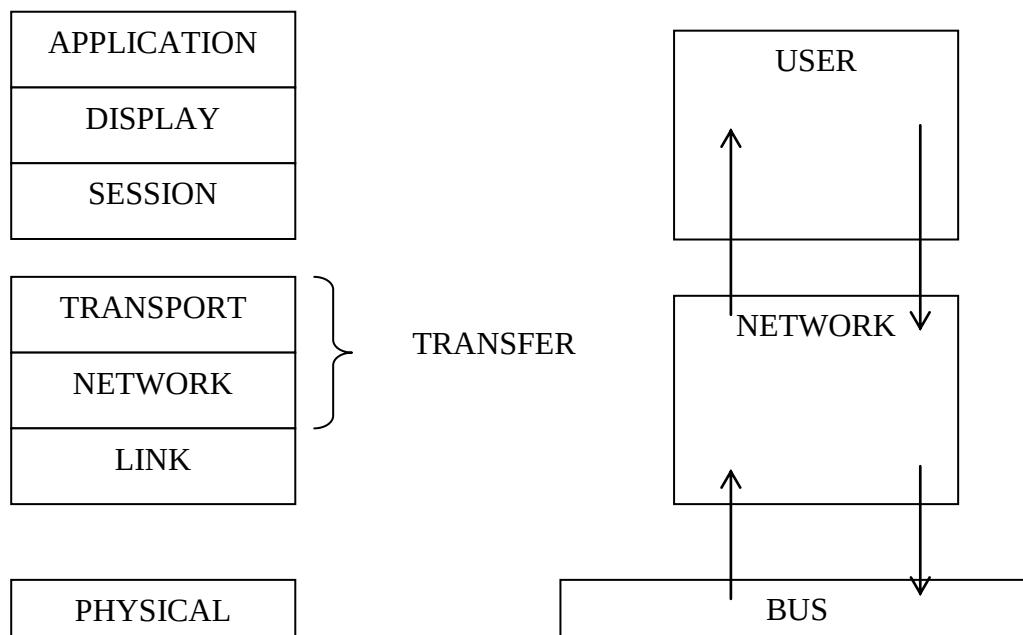
From a functional perspective, an item of equipment is divided into two section:

- One section to ensure application processing
- A second to ensure communications functions



In comparison to the ISO OSI reference model, we obtain the following equivalence:

ISO OSI reference model



The level associated with application processing is called the **USER** level. It corresponds to the three layers labeled APPLICATION, DISPLAY and SESSION in the ISO OSI reference model.

The level used to ensure the bus coupling function is called the **NETWORK** level. This level is composed of the following two layers:

- the **LINK** layer
- the **TRANSFER** layer

The **LINK** layer corresponds to the layer defined in the ISO OSI reference model. Part of the **TRANSFER** layer's functions correspond to the functions of the **TRANSPORT** and **NETWORK** layers of the ISO OSI reference model, the others were added in order to satisfy needs specific to real-time transmission systems comprised of interconnected buses.

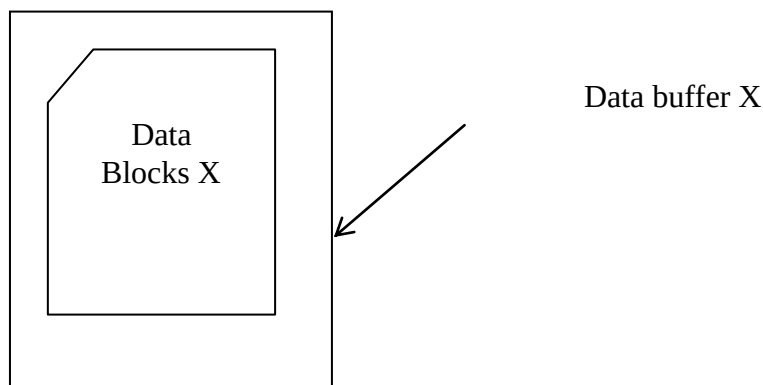
Furthermore, each item of equipment is connected to the transmission **medium** associated with the **BUS** level. This level corresponds approximately to the **PHYSICAL** layer of the ISO OSI reference model.

This breakdown reveals the existence of two logical interfaces:

- The **USER/NETWORK** interface
- The **NETWORK/BUS** interface

5.1.1.1.3 Buffer and Data Block

Different data entities are handled at each level, and possibly at each layer. The **Data Buffer X** designates the container associated with level X or the corresponding layer X. The contents of the Data Buffer is called **Data Block X**.



The size of these buffers and their contents is expressed in 16-bit **words**.

5.1.1.1.4 Protocol

The description of the exchange modes of the Data Blocks X corresponds to the protocol definition of the corresponding level or layer X.

5.1.1.1.5 Equipment hardware

An item of equipment, from a hardware perspective, is divided up into two units. The first unit is the **Host**. The second is the **Network Interface Unit**, commonly known as the terminal.

The distribution of communications functions between these two units is a result of implementation choices. Their distribution may contradict principles defined in the “System objectives” paragraph on page 10.

It is consequently indispensable to take immediate measures to ensure that processing carried out specifically for communications is dissociated at the equipment level from processing relating to the functions that the equipment performs.

It is also recommended that the host and the Network Interface Unit are physically separate (coupling hardware concept).

5.1.1.2 General description of the USER level

5.1.1.2.1 User Data Buffer

A **User Data Buffer (UDBu)** is associated with the User level and contains the data entity exchanged at the USER/NETWORK interface known as the **User Data Block (UDB)**.

User Data Buffer is not limited in size. It should in theory be large enough to contain any User Data Block

5.1.1.2.2 User Data Block

5.1.1.2.2.1 Purpose

The User Data Block is comprised of input and output data from the application processing.

The User Data Block is defined to exchange the same data entity at the USER/NETWORK interface, regardless of the type of processing used.

Two types of processing have been identified.

- Type 1: for a given processing only a single data group is defined to check a minimum set of common criteria which enabled the data to be grouped. The User Data Block corresponds to this data group. It is contained within the associated User Data Buffer.
- Type 2: for a given processing several data groups are defined following the same rules as those described previously. Only one of these groups is produced during a given time lapse. As a result, each of these groups corresponds to a different User Data Block. All groups are contained in the same User Data Buffer, each group at different moments. Each of the User Data Blocks has its own size, whereas the size of the User Data Buffer is fixed. **Filler words** are then used to complete the data. These filler words, also known as **STUFFING** are data words with a value which is not interpreted by the receiver, and can thus have any value.

This method is known as the wagonnet technique and the data is known as **wagon-based**

All input and output data for a processing operation is known as the **Application Data Block (ADB)**. In the event of a Type 2 processing operation, the Application Data Block is known as a **wagonnet**. It is contained in a **wagon**.

5.1.1.2.2.2 Composition

A User Data Block comprises:

- an Application Data Block (ADB)
- an Application Control Block (ACB), optional
- filler words, optional

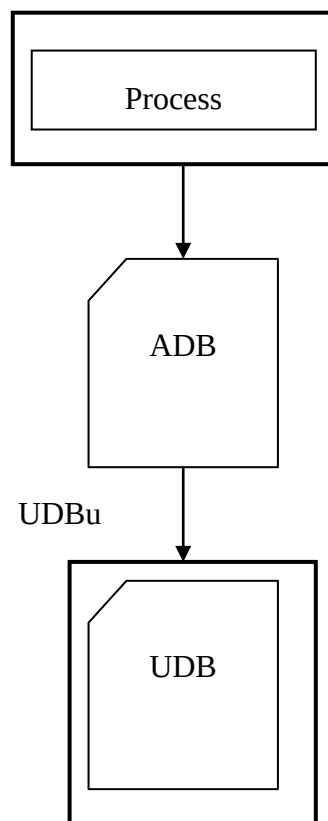
The Application Control Block multiplexes, if necessary, the data at the USER level. The technique used is known as **USER multiplexing**.

The Application Control Block comprises monitoring data for the identification of data when the wagonnet technique is being used.

Two cases must be taken into account:

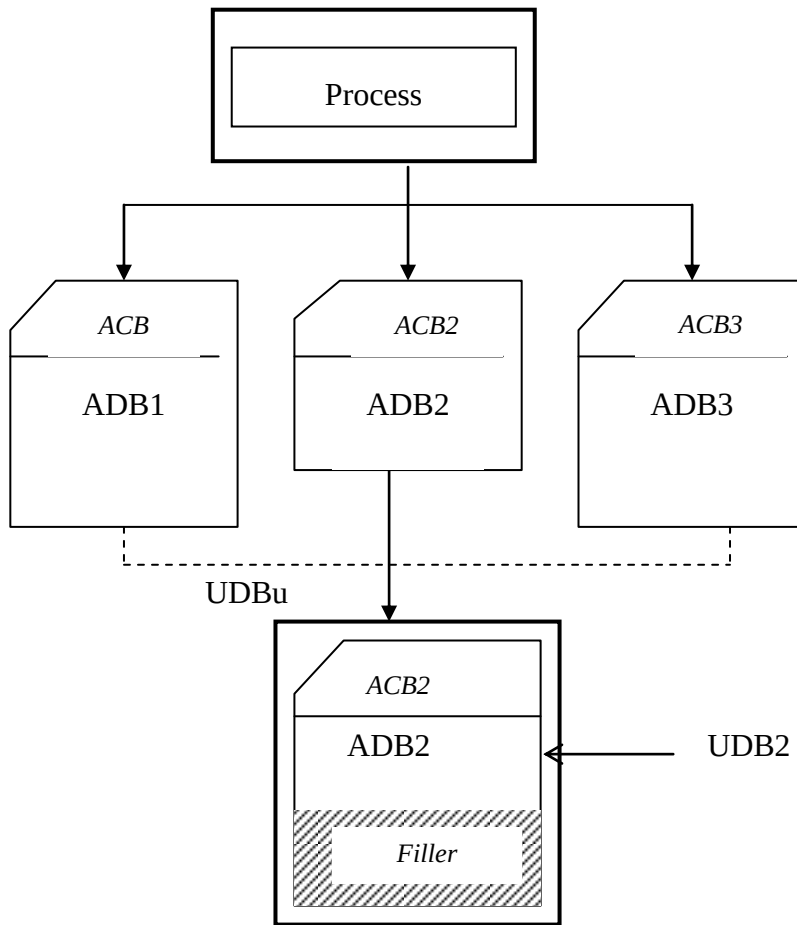
1. No User multiplexing

- **User Data Block = Application Data Block**, The data is not wagon-based :



2. Implementation of User multiplexing

- **User Data Block = Application Control Block + Application Data Block**, the data is wagon-based. The Application Control Block comprises two types of data :
 - **USER multiplexing data**, which corresponds to the **wagon** identifier.
 - **Useful length USER data**, which corresponds to the size of the Application Data Block, or to the size of the wagonnet.



5.1.1.3 General description of the NETWORK level

5.1.1.3.1 Data Link Buffer

A **Data Link Buffer (DLBu)** is associated with the NETWORK level. It contains the data entity exchanged at the NETWORK/BUS interface known as the **Data Link Block (DLB)**.

5.1.1.3.2 Data Link Block

5.1.1.3.2.1 Purpose

The Data Link Block is the data entity transmitted over the bus when a message is sent.

5.1.1.3.2.2 Limitations due to the MIL-STD-1553B protocol

The size of a Data Link Block is less than or equal to 32 16-bit words. This justifies using a segmentation mechanism for an Application Data Block larger than 32 words.

However, 'traditional' use of the MIL-STD-1553B only allows the definition of 30 different Data Link Blocks for each transmission direction (transmission, reception). The sub-address field is encoded in 5 bits and the decimal values 0 to 31 are reserved for mode codes. Nevertheless, certain components control two tables in reception. This enables one to differentiate between messages affecting a single receiver and broadcast messages and to increase the number of Data Link blocks received as a result.

→ A notable increase in the transmission and reception capacities for a given piece of equipment requires the implementation of a sub-address multiplexing mechanism.

5.1.1.3.2.3 Composition

A Data Link Block is comprised of:

- a Data Transfer Block (DTB)
- a Transfer Control Block (TCB), optional
- filler words, optional

The **Data Transfer Block** corresponds entirely or partly to the User Data Block.

The Data Transfer Block performs, if need be, sub-address multiplexing at the NETWORK level.

Sub-address multiplexing and NETWORK segmentation are two functions which are not provided for in the MIL-STD-1553B protocol. These functions are performed by the

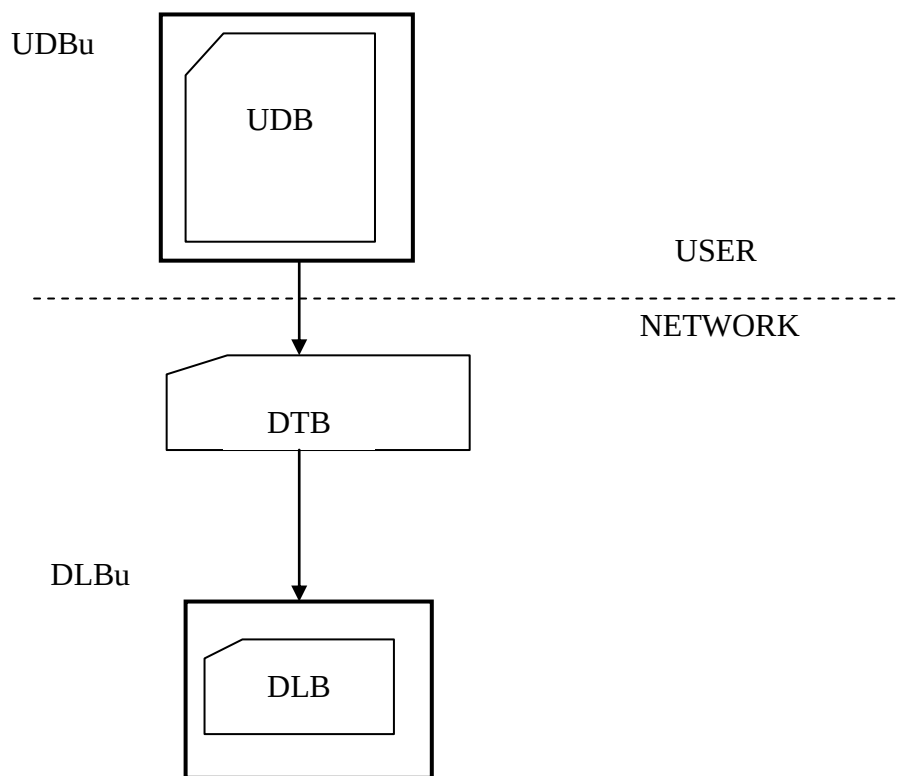
TRANSFER layer. The data required for their performance comprises the **Transfer Control Block (TCB)**.

Filler words are data words that are not processed at USER level during the reception of a message. These filler words, in accordance with coding rules, are forced to 0 by the transmitter.

Several cases are to be taken into account depending on whether or not one implements

1. No sub-addressing and no segmentation

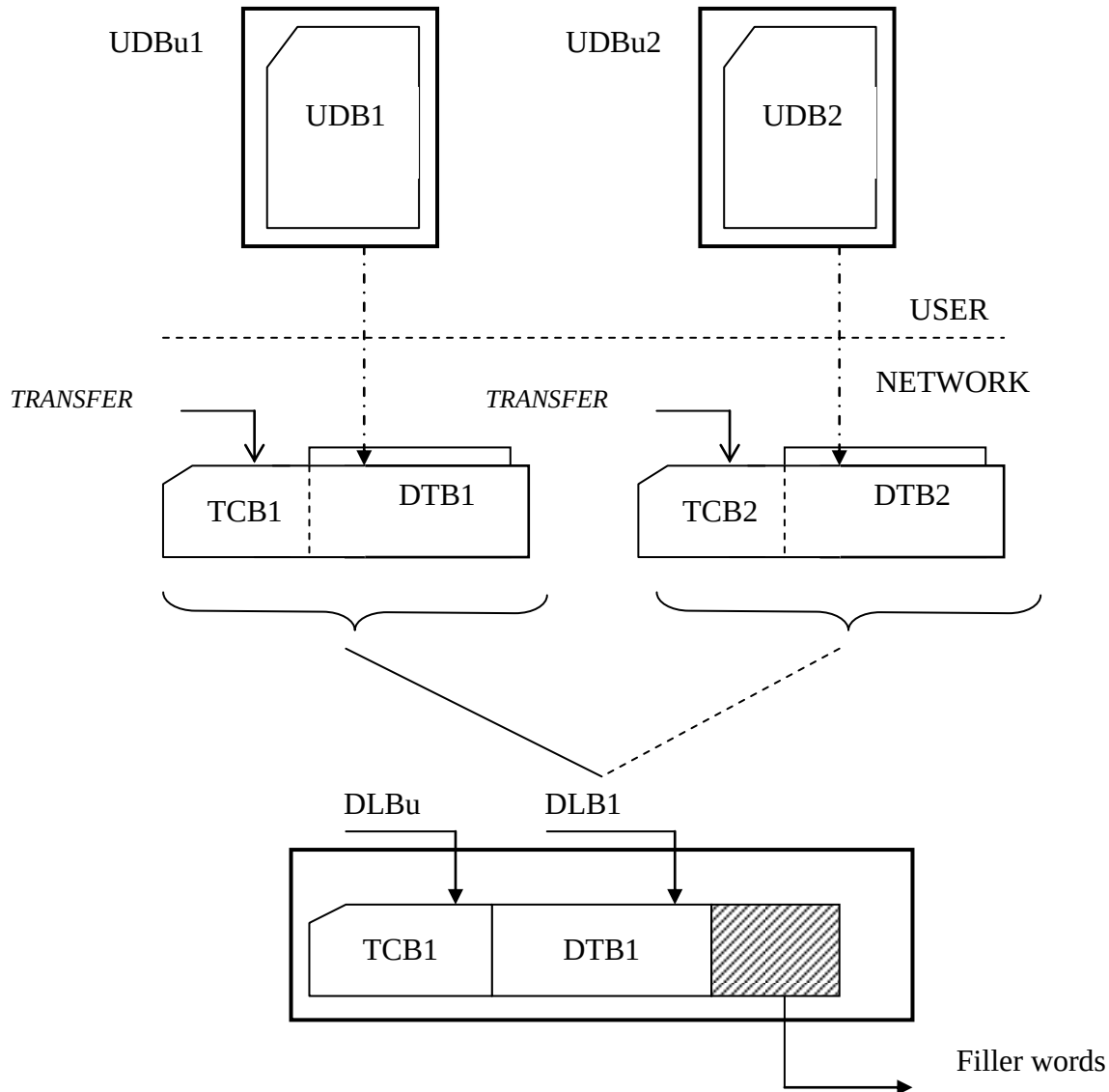
- **Data Link Block = Data Transfer Block**, the size of the Data Link Buffer is sufficient to host the entire User Data Block (DLBu = UDB) :



2. Sub-address extension mechanism and no segmentation

- **Data Link Block = Transfer Control Block + Data Transfer Block + (filler)**: the sub-address multiplexing is used and the size of the Data Link Buffer is sufficient to host each of the User Data Blocks (DTB = UDB(i)).
- The Transfer Control Block contains:
 - ♦ NETWORK multiplexing data, which corresponds to the sub-address extension.

- ◆ Useful length NETWORK data, which corresponds to the size of the Data Transfer Block conveyed.
- The Data Transfer Block is known as a container. It is contained in a “trailer” :



Note: The notable difference between wagon basis (user multiplexing) and Sub-address extension (network multiplexing) is the volume of memory used. In the case of wagon basis, a single User Data Buffer is defined. Its size enables it to contain the largest defined Data Application Block.

3. Segmentation mechanism and no sub-addressing

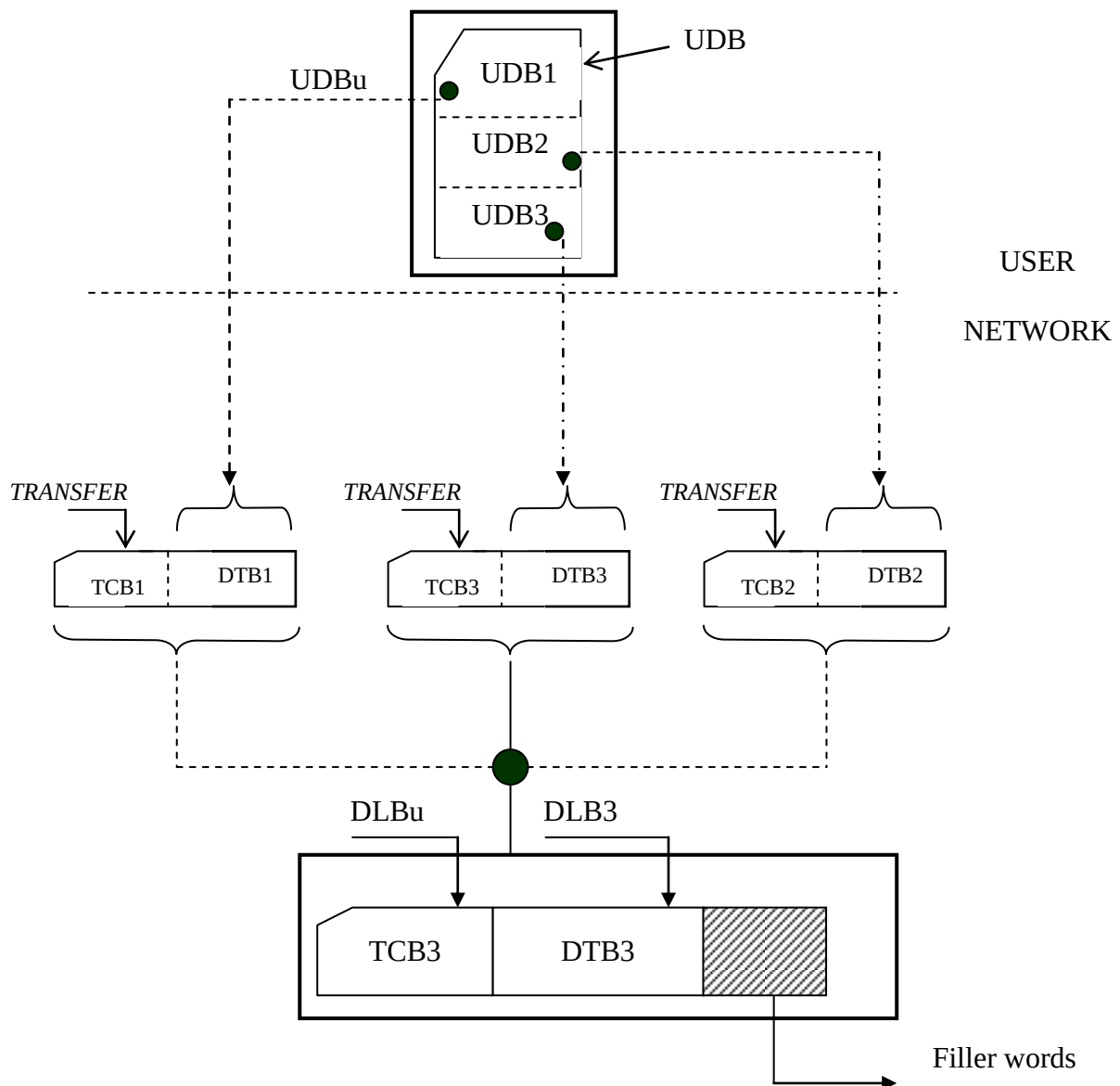
- **Data Link Block = Transfer Control Block + Data Transfer Block + (filler):** the size of the Data Link Buffer is not sufficient to host the entire User Data Block. The User Data Block is thus **segmented**. Two cases are envisaged for this **segmentation**, with respect to optional User multiplexing :

a) No additional User multiplexing (no additional wagon mechanism)

A Data Transfer Block is made to correspond to each segment, known as a **basic user data block (udb)**, when no USER multiplexing has been implemented (DTB = udb).

Note: In this case the filler words are included in the definition of the DLB since the size of the messages is fixed in the case of a centrally controlled system.

SIMPLE SEGMENTATION



The Transfer Control Block, contains:

- NETWORK multiplexing data intended to identify the segment transmitted.
- Useful length NETWORK data, which corresponds to the size of the segment. This data is only useful when the final segment is transmitted, because its size is generally less than the size of all the preceding segments.

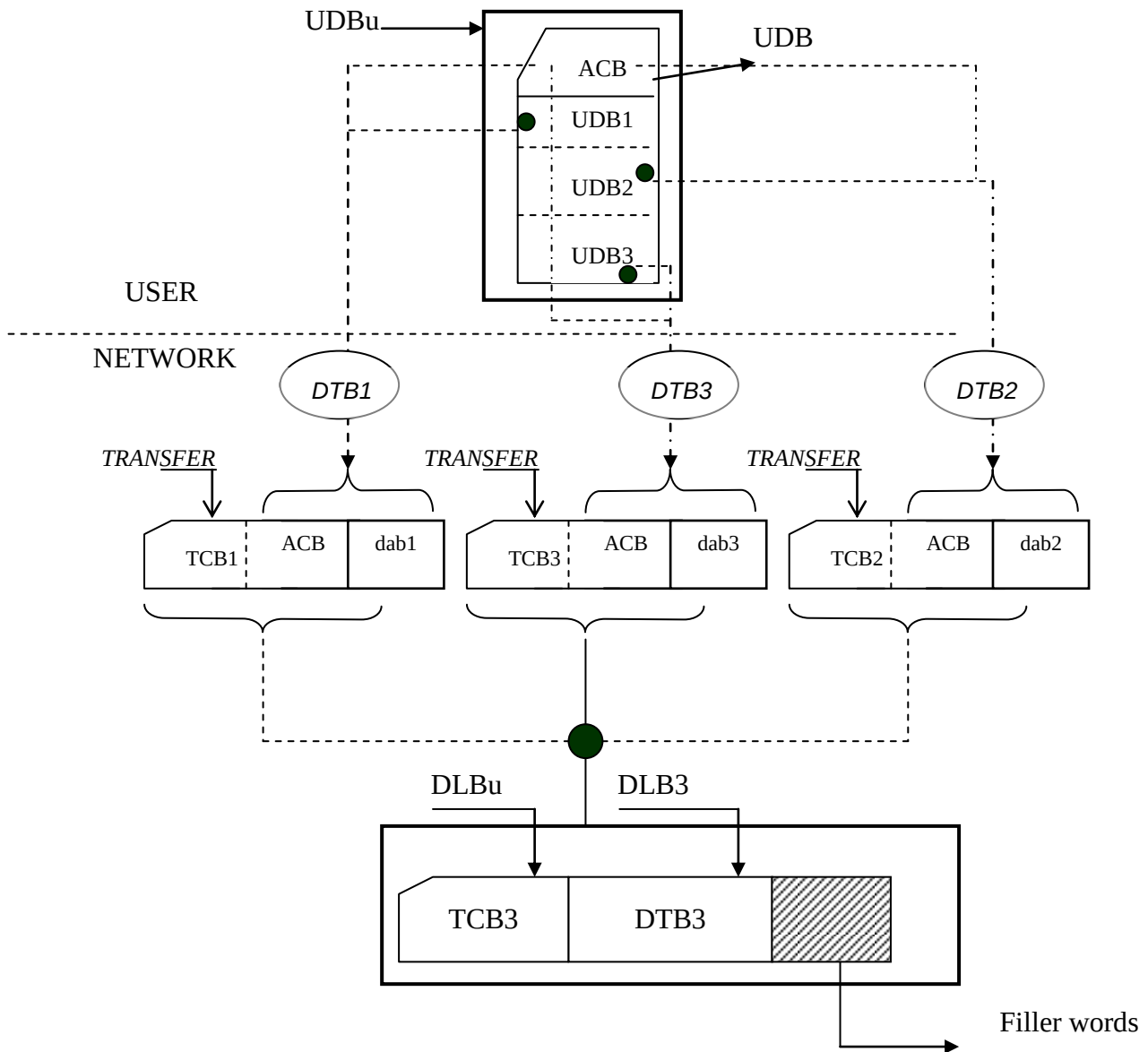
b) additional User multiplexing (additional wagon mechanism)

In the event that USER multiplexing is implemented, a Data Transfer Block includes a basic data user block and an Application Control Block. This facilitates identification and characterization of the wagonnet (DTB = ACB + adb). The segmentation does not concern the Application Control Block. A **basic application data block (adb)** corresponds to a segment of the Data Application Block.

The Transfer Control Block, contains:

- NETWORK multiplexing data intended to identify the segment transmitted.
- Useful length NETWORK data, which corresponds to the size of the segment. This data is only useful when the final segment is transmitted, because its size is generally less than the size of all the preceding segments.

SEGMENTED WAGON



4. Both Segmentation and sub-addressing mechanisms

The Transfer Control Block may contain either or both types of the following network multiplexing data:

- sub-address extension data
- segmentation data

Sub-address extension and segmentation could thus **theoretically** be used simultaneously. Nevertheless, due to Aircraft implementation, the two mechanisms will be exclusive (refer to paragraph 5.1.3.2).

5.1.1.4 General description of the BUS level

5.1.1.4.1 Purpose

The purpose of the BUS level is to send messages over a medium.

5.1.1.4.2 Message

5.1.1.4.2.1 Purpose

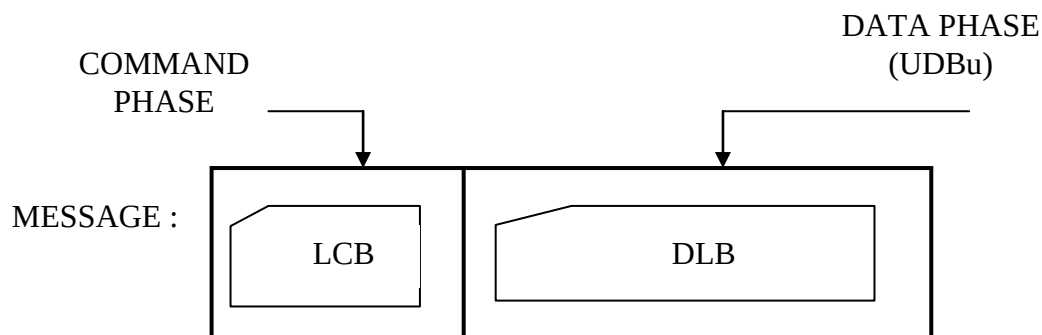
The message is used to carry a Data Network Block which corresponds completely or partially to an Application Data Block.

This Data Network Block must be identified uniquely by the Network Control Block.

5.1.1.4.2.2 Structure

- **Physical description :**

A message comprises a command phase containing a Link Control Block (LCB), and a data phase containing a Data Link Block.

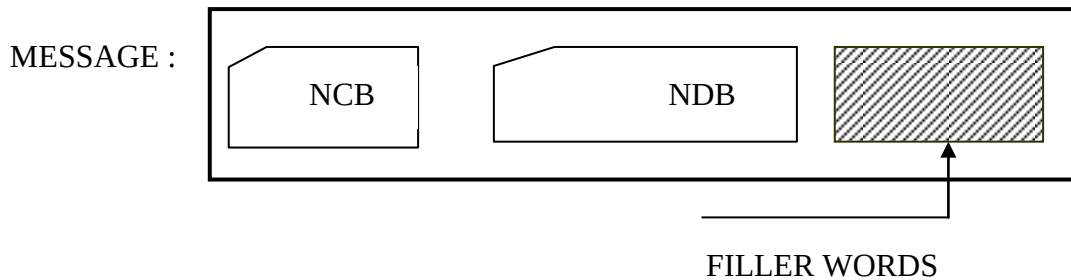


In the MIL-STD-1553B protocol, the command phase corresponds to **command words** (transmission command word, reception command word) and to **status words** (transmission status word, reception status word). The data phase comprises **data words** only. **Filler words** are a special type of data word.

The maximum size of the Data Link Block is limited by the size of the data phase. The data phase corresponds to the Data Link Buffer.

- **Functional Description :**

In this case, it is considered that each message is made up of a **Network Control Block (NCB)**, a **Network Data Block (NDB)** and possibly **filler words**.



The correspondences between the different control blocks (NCB, TCB, LCB) and between the different Data Blocks are dealt with in the paragraph “General description of the composition of messages” on page 30.

5.1.1.4.2.3 Type

All messages transmitted on a bus can be divided into 2 categories:

- **Operational messages** which are used to exchange operational data between WDNS equipment.
- **Service messages** used by dialog protocols which facilitate the implementation, monitoring and control of data exchange between items of equipment.

The definition of operational messages depends for the most part exclusively on the operational functions to be carried out. However, service messages used for dialog protocols on aircraft are standardized. A list of these protocols appears in the paragraph entitled “Messages associated with predefined sub-addresses” on page 56. This list is made up of mode codes used amongst those defined in the MIL-STD-1553B protocol and of a certain number of data messages simplified for use at transmission system level.

5.1.1.4.2.4 Characteristics

The main characteristic of a message is the **size** of the data phase. Its maximum value authorized by the MIL-STD-1553B standard is 32 16-bit words.

Every message has a fixed size. At BUS level, the size of the data phase for a data message is between 1 and 32 words. Data Link Blocks and the data phase thus have the same size which corresponds to the size of the “word count” entered in the command words.

Nevertheless, in order to facilitate the modification of messages at NETWORK level, the memory reserves must be such that the size of the Data Link Buffers for all messages can increase up to a size of 32 words.

5.1.2 General description of the composition of messages

5.1.2.1 Simple use of each mechanism

If it is borne in mind that each mechanism is used individually, the following cases can thus be determined:

1. First case

Network Control Block = Link Control Block

Data Link Block = Network Data Block = Data Transfer Block

The MIL-STD-1553B protocol is used in a traditional manner, i.e. without addition of a multiplexing mechanism (sub-address extension or wagonnet) or segmentation. For a given message, the Network Control Block, reduced to the Link Control Block, uniquely identifies the Data Network Block contained in the data phase. The Data Link Block and the Data Transfer Block are grouped together.

2. Second case

Network Control Block = Link Control Block + Transfer Control Block

Data Link Block = Transfer Control Block + Network Data Block + (filler) = Transfer Control Block + Data Transfer Block + (filler)

A sub-address multiplexing or segmentation mechanism is used in addition to the MIL-STD-1553B protocol. The Network Data Block contained in the data phase is, for any message, identified uniquely by the Network Control Block. The Link Control Block can no longer ensure that the identification is unique. Additional identification is carried out by the Transfer Control Block, the Network Data Block and the Transfer Data Block collectively.

3. Third case

Network Control Block = Link Control Block + Application Control Block

Data Link Block = Application Control Block + Network Data Block + (filler) = Data Transfer Block

When the wagonnet technique is used, the Network Data Block contained in the data phase is identified uniquely by the Network Control Block. The Link Control Block can no longer ensure that the identification is unique. Additional identification is carried out by the Application Control Block.

5.1.2.2 Definition of the Additional Control Block

The **Additional Control Block (AdCB)** is composed of additional data necessary for the identification and the characterization of the Network Data Block carried.

Depending on the situation, the following are available:

- **Additional Control Block = Application Control Block:** USER multiplexing
- **Additional Control Block = Transfer Control Block:** multiplexing and/or NETWORK segmentation

5.1.3 Application to Dassault Aviation A/C

5.1.3.1 Methods used

The methods used are the following:

- At NETWORK level, sub-address multiplexing or segmentation.
- At USER level, the wagonnet technique.

The combined use of these different mechanisms is **prohibited**.

The additional control data necessary for the identification and the characterization of the Network Data Block are contained, irrespective of the method used, in the two first words of the data phase, known as the **header**.

The selection of methods to implement depends on the dialog bus which must be ensured at each piece of equipment.

Choosing one of the 3 methods depends on the sub-address capacity available and the size of the User Data Block.

5.1.3.2 Limitations

Simultaneous use of sub-address multiplexing mechanisms wagonnets is **not possible**. Sub-address extension and segmentation mechanisms can also not be used together.

Taking into account the size of the header, the transfer of User Data Blocks with a size in excess of 32 words requires the implementation of a segmentation mechanism.

A single Network Data Block is carried per message. Thus, no mixture of data can take place (wagon-based or non-wagon based) in a single Link Data Block.

The header contains two other data items:

- **Multiplexing data** which is used to identify the Network Data Block carried by complementing the addressing data contained in the command phase of the message.
- **Useful length data** which is used to determine the size of the Network Data Block.

The MIL-STD-1553B protocol is to be used in the “traditional” manner. Sub-address multiplexing is only to be implemented when necessary.

Except in the event of a transmission error, a single Network Data Block cannot be sent more than once per short cycle.

1. Without multiplexing (USER OR NETWORK), the same receiving sub-address can only be used once per short cycle.
2. With multiplexing (USER OR NETWORK), the two same sub-addresses can only occur in a short cycle on condition that the multiplexing data items are different.

5.1.3.3 General composition of the Link Data Block

The methods used and the associated limits require the following Link Data Block structure:

Data Link Block = (Additional Control Block) + Network Data Block + (filler)

The Additional Control Block is optional. It contains control data at NETWORK level or USER level.

The filler words only exist if the mechanisms used require the setting of an Additional Control Block. Their existence results from the fact that each Data Block has its own particular size. Each Data Block is contained in the same data phase with a fixed size between 1 and 32 words. This size is entered into the Link Control Block contained in the command phase of the corresponding message. It is equal to the size of the data phase necessary for the transfer for a given message of the larger Network Data Block. The filler words are used to fill in the last words of the data phase for all other Network Data Blocks sent using this same message.

5.1.3.4 General composition of the Additional Control Block

Using the same message to send several different Network Data Blocks requires that an additional identification be set in the data phase of the message. The data is known as **multiplexing data**.

In addition, the use of multiplexing mechanisms (USER for wagonnets, NETWORK for sub-address multiplexing) requires the use of filler words, that are not to be processed by the receiver(s). It is thus necessary to indicate the useful length for the data phase. This value corresponds to the size of the Network Data Block contained in the data phase. This will cause a **useful length** field to be set in the data phase.

The **Additional Control Block** is composed of these two items of data.

5.1.3.5 Detailed composition of the Link Data Block

The possible cases encountered depend on:

- The size of the User Data Block
- The identification mode of the Network Data Blocks

A User Data Block is said to be of **limited size** if it can be transferred by sending a message once. In all other cases, it is considered to be of **unlimited size**.

Depending on the mechanism used, the maximum size used to distinguish between “limited size” and “unlimited size” is different:

- If no multiplexing mechanism is used, the User Data Block is said to be of limited size if its size, which is equal to the size of the corresponding Application Data Block, is less than or equal to 32 words.
- If a NETWORK multiplexing mechanism is used, the User Data Block is considered to be of limited size if its size, which is also the size of the corresponding Application Data Block, is less than or equal to 30 words. Two data words must be provided for the Additional Control Block.

If a USER multiplexing mechanism is used, the User Data Block is said to be of limited size if its size is less than or equal to 32 words. In this case this size does not correspond to the size of the associated Application Data Block because the two data words of the Additional Control Block must be taken into account. These correspond to the Application Control Block.

The possible cases identified are the following:

- No multiplexing (USER or NETWORK) and limited User Data Block size
- No multiplexing (USER or NETWORK) and unlimited User Data Block size
- USER multiplexing (wagon based) and limited User Data Block
- NETWORK multiplexing (sub-addressing extension) and limited UDB size

5.1.3.5.1 No multiplexing and limited User Data Block size

The MIL-STD-1553B protocol is used in a traditional manner. The User Data Block is transmitted all at once by sending a message.

The following is thus obtained:

Data Link Block = Network Data Block = Data Transfer Block = User Data Block = Application Data Block

Network Control Block = Link Control Block

The identification of the Network Data Block is carried out through addressing data contained in the basic command phase stipulated in the MIL-STD-1553B protocol.

The size of the Network Data Block corresponds to the value of the word count contained in the basic command phase stipulated in the MIL-STD-1553B protocol.

5.1.3.5.2 No multiplexing and unlimited User Data Block size

The User Data Block is transmitted several times by sending the same message several times. This mechanism is called segmentation.

Each segment corresponds to the Network Data Block contained in the data phase of each of the messages.

The following is thus obtained:

User Data Block = Σ user data block(j) = Application Data Block

Data Transfer Block(j) = Network Data Block(j) = user data block(j)

Network Control Block (j) = Link Control Block + Transfer Control Block(j)

“j” relates to a segment.

The identification, at the level of a given bus, of each of the segments is carried out by applying the addressing data contained in the command phase of the message used. This data is complemented by the multiplexing data contained in the header (segment identifier).

The length of each segment is given by the value of the useful length data contained in the header.

5.1.3.5.3 USER multiplexing and limited User Data Block size

Each of the User Data Blocks produced by the associated processing operation is transmitted all at once by sending the same message.

The following is thus obtained:

User Data Block(i) = Application Control Block(i) + Application Data Block(i)

Data Link Block(i) = Application Control Block(i) + Network Data Block(i) + filler (i)

Network Data Block(i) = Application Data Block(i)

Network Control Block(i) = Link Control Block + Application Control Block(i) = Link Control Block + Additional Control Block(i)

“i” relates to the different User Data Blocks.

As a result, the identification and the size of the User Data Block thus carried are achieved using the USER level additional control data making up the Application Data Block.

- wagonnet identifier (USER multiplexing data)
- the size of the wagonnet (USER useful length data)

5.1.3.5.4 NETWORK multiplexing and limited User Data Block size

Each of the User Data Blocks produced by one of the processing operations considered is transmitted all at once by sending the same message.

The following is thus obtained:

Data Link Block(i) = Transfer Control Block(i) + Network Data Block(i) + filler (i)

Network Data Block(i) = User Data Block(i)

Network Control Block(i) = Link Control Block + Transfer Control Block (i) = Link Control Block + Additional Control Block(i)

“i” relates to the different User Data Blocks.

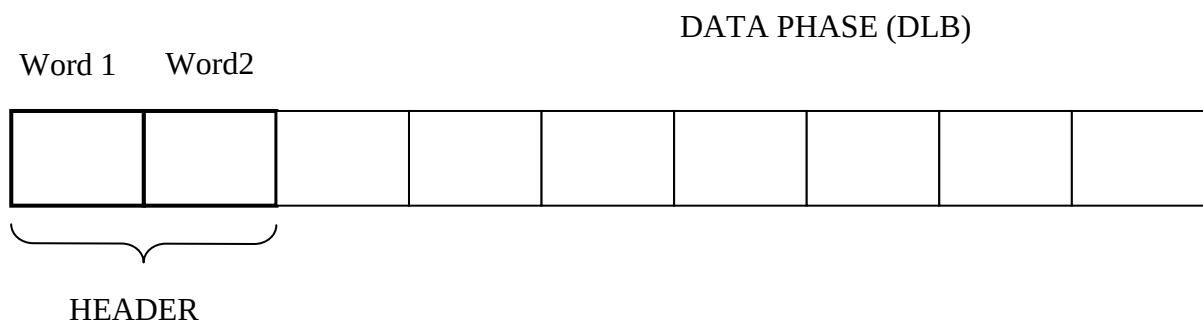
The identification and the size of the User Data Blocks thus carried are achieved using the NETWORK level additional control data making up the Transfer Control Block. The items of data are

- NETWORK multiplexing data
- NETWORK useful length data

5.1.3.6 Structure of the data phase

5.1.3.6.1 Definition of a header

In the event that additional identification of the Network Data Blocks is used, the first two words of the data phase make up the **header**. This header contains the Additional Control Block.

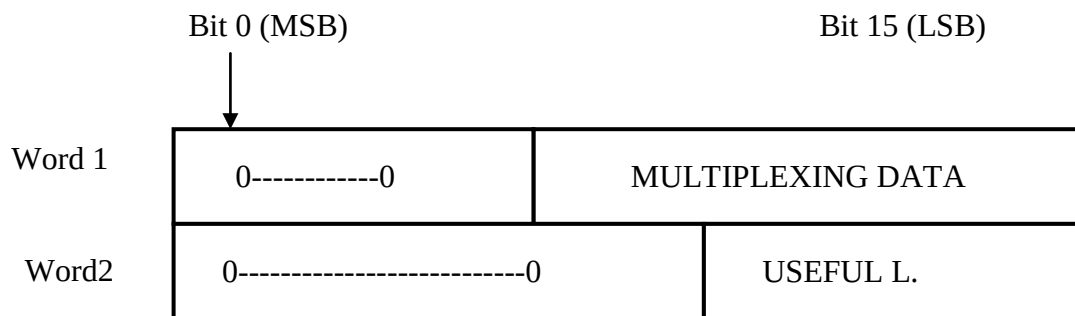


Whatever the mechanism used, the header contains multiplexing data and useful length data. No distinction is therefore possible from the point of view of the bus, this being the aim sought. With regard to equipment, the distinction occurs at the NETWORK level, through prior knowledge of multiplexed sub-addresses.

5.1.3.6.2 Header coding

Given that the Transfer and Application Control Blocks used for sub-address extension and wagonisation are composed of the same parameters, the coding for the header is the following:

- The first word of the header contains the “multiplexing data” parameter. This is encoded in 8 right-justified bits (bits 15 to 8, imsb: bit 8, ilsb: bit 15), in compliance with DGT document no. 81070 “General coding rules”. The bits not used are forced to 0. This multiplexing data will never be zero.
- The second word of the header contains the “useful length” parameter, which is expressed in words. This header is encoded in 5 right-justified bits (bits 15 to 11, imsb: bit 11, ilsb: bit 15), in compliance with DGT document no. 81070 “General coding rules”. The bits not used are forced to 0.



The Transfer Control Block used in the event of segmentation must comply with the description in document GAM-T-110.

5.2 Exchange protocols

5.2.1 Overview

The purpose of the data transfer service is to transmit application data between the processing operations divided amongst the different items of equipment.

Depending on the size of the Application Data Block, simply sending a message may not ensure the transmission of an Application Data Block.

The transmission of an Application Data Block corresponds to an **APPLICATION exchange**. All the rules allowing this transmission constitute the **APPLICATION protocol**.

The transmission of a message corresponds to a **LINK exchange**. All the rules allowing this transmission constitute the **LINK protocol**.

Thus, an APPLICATION exchange may implement several LINK exchanges. Nevertheless, in most cases, a LINK exchange corresponds to an APPLICATION exchange.

5.2.2 APPLICATION exchange protocol

5.2.2.1 Definition of an APPLICATION exchange

An APPLICATION exchange is defined by:

- Its type and associated characteristics
- The processing/communication synchronization mode used
- The data production/consumption mechanism used

5.2.2.2 Type and characteristics of an APPLICATION exchange

5.2.2.2.1 Time-division aspect

The transmission of an Application Data Block can be periodic or a-periodic. Two types of APPLICATION exchange are thus to be considered:

- A periodic APPLICATION exchange
- An a-periodic APPLICATION exchange

5.2.2.2.2 Security aspects

5.2.2.2.2.1 Expression of need

In certain cases, the occasional loss of data does not have serious consequences. This is the case for the majority of data transmitted periodically. However, the loss of certain data transmitted randomly or even the occasional loss of certain data transmitted periodically, is not acceptable.

Two types of exchange are to be considered:

- Secure APPLICATION exchanges
- Non-secure APPLICATION exchanges

5.2.2.2.2.2 Implementation

Security can be ensured in two ways:

By the retransmission of all or part of the data making up the Application Data Block. This implies the retransmission of one or more messages used to transfer the Block. Depending on the size of the Block, either the NETWORK level alone is concerned or the NETWORK and USER levels. The security concerns the transfer of data.

By sending additional control data (i.e. a Cyclical Redundancy Code added to the data which makes up the Application Data Block). This method falls within the scope of the USER level. The security concerns the data.

To be efficient, an exchange must be made secure at the USER level.

The simple retransmission of messages is carried out principally for a-periodic exchanges. This is based on the fact that the loss of a data sample is not of major importance at the periodic exchange level, though it might be at the a-periodic level.

5.2.2.3 USER/NETWORK Synchronization

5.2.2.3.1 Analysis of synchronization mechanisms

5.2.2.3.1.1 Reminder

With regard to the functioning of a terminal (Network Interface Unit), only transmission mechanism for messages exists at NETWORK/BUS interface level. However, several possibilities are available for the USER/NETWORK interface to activate transmission or indicate reception of the USER DATA BLOCK. These possibilities ensure, or do not ensure, the synchronization of processing and communications functions for a piece of equipment.

5.2.2.3.2 Methods used

This paragraph is concerned only with processing operations concerning to data exchanges performed via the MIL-STD-1553B bus.

5.2.2.3.2.1 Cyclical operations: nominal mode

In general, the duration of a basic **cyclical operation** is 3 cycles which are equal in duration to the activation period of the operation.

cycle n reception of processing input data

cycle n+1 execution of the processing operation

cycle n+2 transmission of processing output data

The advantage of this sequencing is that any constraint relating to the setting of the message in the frame is avoided. This presupposes however that the transmission and processing cycles are phase and frequency synchronized.

This is the basic sequencing mode used. In this case, the reception of messages containing the input data is not indicated at USER level. Moreover, the messages carrying output data are transmitted at the initiative of the controller. The only necessary reference, both with regard to the application processing and the bus transmission, is the beginning of the cycle.

5.2.2.3.2.2 Cyclic operations: exceptional mode

For reasons relating to the optimization of response time, the duration of a basic processing operation may be reduced to two, and even one cycle. This presupposes an indication at USER level of the reception of the User Data Block containing input and/or output data at the same level as the User Data Block containing output data.

These functionality, although used specifically, are to be provided for all items of equipment.

5.2.2.3.2.3 Processing on request

Processing **on request** operations are operations that can be activated on reception of data transferred through the transmission of messages on request. Input and processing data may thus be received in the same cycle. Leaving a margin for error, the duration of a cycle corresponds to the response time specified for the transmission of messages on request.

5.2.2.4 Consumption/production of data

5.2.2.4.1 Overview of the mechanisms

At the USER level, two consumption modes for the User Data Block are to be taken into account for both reception and transmission. These modes are **sampling** mode and **flow** mode.

Sampling mode enables the use of the latest Data User Blocks at the risk of losing older ones. During transmission, the latest Data User Block(s) produced deletes the previous ones. During reception the latest Data User Block(s) received deletes the previous ones.

Flow mode enables the receiver(s) to use all the samples, which causes the retransmission of the same User Data Block if the speed of production at the transmitter level is greater than the speed of consumption of the receiver, taking into account available memory resources. During transmission, this mechanism may be represented by a queue used to stock Data User Blocks awaiting transmission. During reception, this mechanism may be represented by a queue used to stock Application Data Blocks awaiting processing.

5.2.2.4.2 Use

Sampling mode is the mode most commonly used.

5.2.3 LINK exchange protocols

5.2.3.1 Definition of a LINK exchange

A **LINK exchange** involves the transmission of a message. The exchange is dynamic while a message is of a static nature.

An exchange is defined by its type and its associated characteristics.

It should be noted that the MIL-STD-1553B is a LINK level protocol. It is used to transmit a message.

5.2.3.2 Type and characteristics of a LINK exchange

5.2.3.2.1 Topology of the exchange

The topology of an exchange is the description of the items making up the exchange: the transmitter and the receivers of the message.

When the equipment connected to the bus via a controller is not affected by the exchange (for transmission or reception) the exchange is known as **decentralized**.

Otherwise, the exchange is said to be **centralized**. It is referred to as a **centralized transmitter** or **centralized receiver** depending on whether the equipment connected to the bus via the controller transmits or receives the message. An exchange can only be a centralized receiver when this equipment is the only receiver of the message.

Depending on the number of receivers, the exchange is known as a **point-to-point exchange** or **broadcast exchange**.

Only one receiver is involved in a point-to-point exchange. Several receivers are involved in a broadcast exchange. **Generalized broadcast**, in which all the items of equipment connected to the bus (except the transmitter) are receivers of the message, can be distinguished from **selective broadcast** in which, of the items of equipment connected to the bus, only one subsystem is involved in the message.

5.2.3.2.2 Existence

To the extent that the transmission of certain data is necessary all the time, and other data only partially if certain conditions are fulfilled, **continuous** exchanges can be distinguished from **conditional** exchanges. For any conditional message an **activation condition** must be made explicit at the application level. This condition is specific to the message or common to several messages.

5.2.3.2.3 Periodicity

A data exchange may be:

- **Periodic:** In the absence of error, this exchange corresponds to the systematic activation of the transmission of the same message. The interval of time separating two consecutive activations of the same message is constant and equal to the transmission period of the message.
- **A-periodic:** this exchange corresponds to the transmission of a message for which the time interval separating two consecutive transmissions is random.

5.2.3.2.3.1 Periodic exchange

At digital data transmission system level, the time is divided into cycles. In the case of a centralized control system, the transmission of all messages for all equipment is carried out in relation to a single basic time, which is the basic time for the bus controller. The associated temporal axis is known as a **temporal network axis**.

As a result, a T period **periodic exchange** corresponds to the single transmission of the message per cycle with a duration equal to T. The exchange is then known as **cyclic**.

The basic cycle is known as a **short cycle**.

5.2.3.2.3.2 A-periodic exchange

The **a-periodic exchanges** are commonly called random exchanges.

5.2.3.2.3.3 Activation type

A message can be transmitted either **automatically**, in which case the exchange is known as an **automatic exchange**, or **on request**, in which case the exchange is known as **exchange on request**.

An exchange is said to be automatic if it is transmitted at the initiative of the NETWORK level, thus at the initiative of the controller for a centralized system. It is said to be on request if it is transmitted at the initiative of the USER level of any of the equipment connected to the bus.

5.2.3.3 Cyclic exchange protocol

5.2.3.3.1 Data transmitted cyclically

Data with constantly varying value which is sampled at system level is transmitted cyclically.

5.2.3.3.2 Description of the protocol

No particular protocol is defined for the cyclic exchanges. The temporal network axis is divided into **short cycles**, each with a duration of 20 milliseconds. Eight short cycles are grouped to form a **long cycle** with a duration of 160 milliseconds.

5.2.3.3.3 Definition of the frequencies

The **transmission frequencies** used for cyclic exchanges are 50Hz, 25Hz, 12.5Hz and 6.5Hz.

The data transmission frequency must be selected in agreement with DASSAULT AVIATION, taking into account the accuracy desired and not using a higher frequency to solve response time problems.

5.2.3.4 Random exchange protocol (or exchange protocol on request)

5.2.3.4.1 Protocol description

- **Brief overview of the random message mechanism**

When the application of a remote terminal wishes to transmit or receive messages on request, it alerts the associated transfer layer. This layer stacks up the requests and initiates a counter (or a latency time delay) linked to each request. It allocates the necessary memory resources to the exchange, sets the highest priority request number in the vector word and then sets the “Service Request” bit of the status word to 1. On reception of the polling message, the status word and the vector word are sent. The “Service Request” bit is then set to 0. The transfer layer can then reset the vector word to 0 in the absence of any new request, or, if it wishes to request or transmit a new random message, it can allocate the associated memory resources to the new message with the highest priority, set the number of this new request in the vector word and then set the “Service Request” bit to 1.

After sending a non-zero vector word, and in the event that no refusal message has been transmitted (see the paragraph entitled “Refusal Message” on page 47), the remote terminal must receive the transmission or reception command corresponding to the request.

In the case of a transmission command, the remote terminal coupler transmits data that the application has placed in the transmission buffer. In the case of a command for reception, the remote terminal coupler sets the data received in the reception buffer. At the end of the exchange, the associated transfer layer interrupts the associated counter and informs the application.

If the counter attains the maximum specified value, the latency time delay is exceeded and the application is informed: the connection must be purged, i.e. everything that concerns the request at transfer level must be “forgotten”.

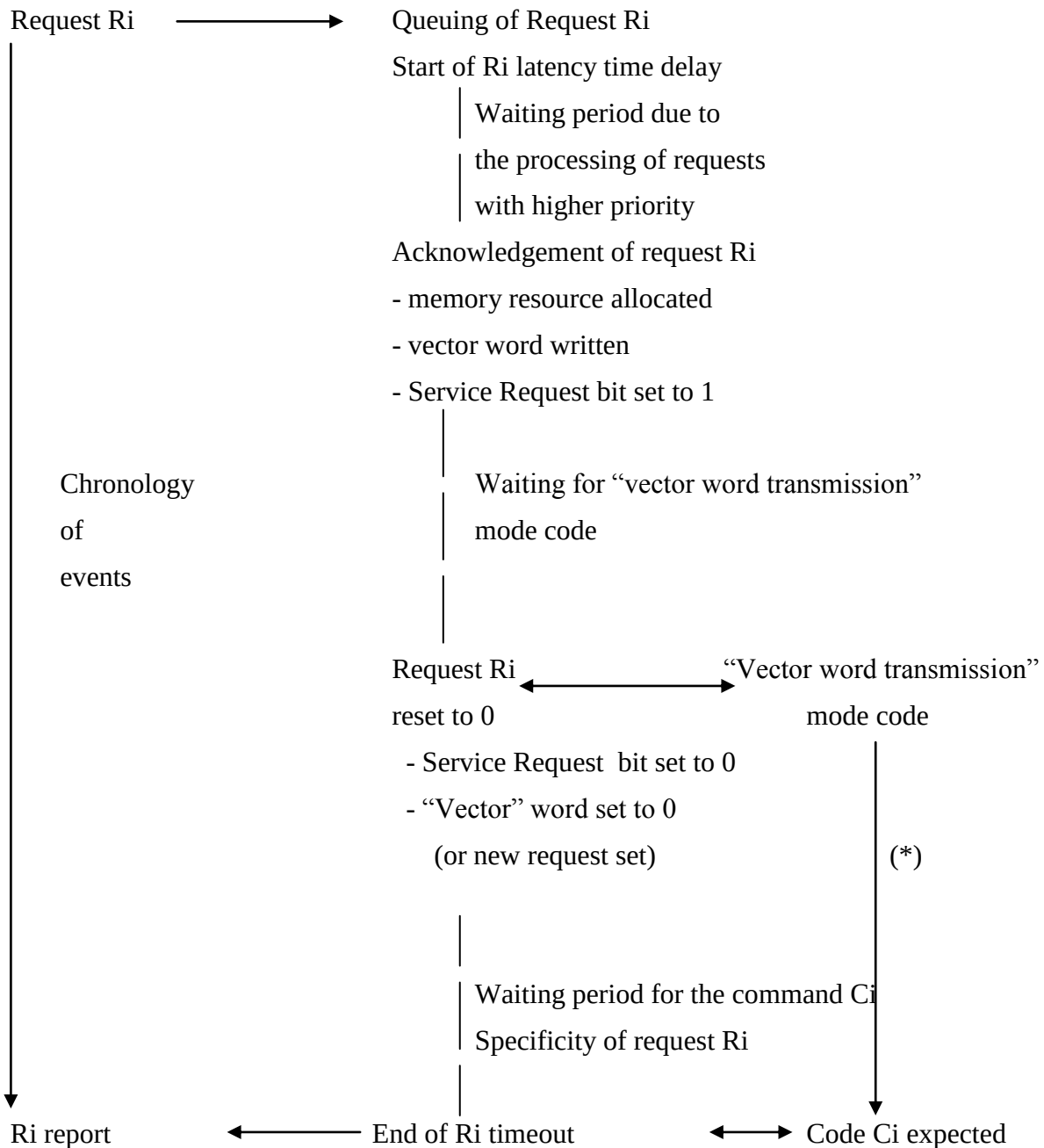
After the latency time delay has been exceeded, the application may abandon or renew the request. The consequences of exceeding the latency time delay must be studied case by case.

The diagram below summarizes all the mechanisms that need to be implemented by the transfer layer so that an exchange on request can be carried out:

APPLICATION

TRANSFER

BUS



(*) Indicates controller response time

Note 1: The expiry of the latency time delay should enable the application to know that its request will not be taken into account by the controller. In the absence of particular specifications at the application level, the value of the latency time delay must be increased to avoid taking into account a mere delay in the transmission of the random data. The value should be around 500msec (approximately 3 cycles long) for any random data, regardless of its priority.

Note 2: The transfer layer of an item of equipment must manage a latency time delay for each request made by the application. A mechanism for the management of the latency time delay may be implemented that is common to all requests (for example, during reception of the request sent by the application, it is possible to store the value of a counter maintained internally from which the latency time delay is exceeded and to check after each increase of the counter whether this value has been reached).

Note 3: A random exchange must involve the use of the following memory management mechanisms:

- Master transmitter exchange:
 - ◆ Transmitting terminal: the flow mechanism¹ is required to ensure that the application can transmit all that it wishes to transmit.
 - ◆ Receiving terminal: the mechanism may be flow¹ or sampled² depending on how the data is used by the application.
- Master receiver exchange:
 - ◆ Transmitting terminal: the sampled mechanism² is necessary so that the transmission buffer can always contain messages that are ready to be sent.
 - ◆ Receiving terminal: the flow mechanism¹ is necessary so that the application can process all the data that it requested.

It should be remembered that for a cyclic exchange, it is necessary to take into account only the most recent data: the sampled mechanism must be implemented for transmitting and receiving terminals.

5.2.3.4.2 Data transmitted randomly

¹In flow mode, no data is lost. Traditional implementation of the flow mechanism leads to the memory being managed as FIFO.

²In sampling mode, only the latest data must be taken into account. Traditional implementation of the sampled mechanism entails the use of a single memory buffer or 2 buffers that are flip-flop managed

The use of random exchanges is feasible for the transmission of discrete data from an event external to the system (keystrokes, appearance of a threat, etc.) or events internal to the system (appearance of failures, etc.). The use of random exchanges is also feasible for the transmission of digital data that does not change (e.g. updating parameters, static characteristics of a target, selector switch status, etc.).

The random exchange protocol will be used systematically for the transmission of such data if there is a gain at the bus load level.

All terminals must be capable of following the transmission protocol for messages on request.

5.2.3.4.3 Exchange request

5.2.3.4.3.1 Identification of exchange requests

Exchange requests are identified in the polling message. This message, which is transmitted on a main line, corresponds to the **vector word transmission** mode code in which the exchange request is identified (no. of the desired request).

The transmission of this request is cyclic and its frequency is specific to each receiving remote terminal.

Vector word coding is described in the paragraph entitled “Logical addressing of Network Data Blocks (NDB)” on page 58. If the content of a vector word is null (zero), this means that the item of equipment polled has no exchange request.

The definition of these different exchange requests is carried out exclusively by DASSAULT AVIATION

5.2.3.4.3.2 Validity of the request

The remote terminal always enters its vector word before validating it by setting the Service Request bit to one.

A remote terminal may modify the contents of the vector word if an exchange request with a higher priority has been indicated by the USER level. In this case, the following sequence must be observed:

1. Reset Service Request to zero
2. Save contents of the vector word
3. Modify vector word
4. Set Service Request bit to one

The **vector word** is systematically **reset to zero** after each of its transmissions. It is recommended that this resetting to zero be carried out just after resetting the Service Request bit to zero.

5.2.3.4.3 Scheduling of the exchange requests

The scheduling of the exchange requests occurs at two levels:

- **At the local level** by the remote terminal which classifies its different requests based on the level of priority accorded to the requests.

At the overall level by the controller which classifies the requests performed by all the remote terminals and the host computer to which it is associated based on the level of priority accorded to the requests.

5.2.3.4.4 Refusal message

Point-to-point and broadcast refusal messages are transmitted exclusively on the same sub-address at any given moment.

5.2.3.4.4.1 Point-to-point refusal

The controller transmits 3 data words to sub-address 3 in point-to-point mode on the main line. This is a random type message with a priority specific to each of the receiving remote terminals.

The controller transmits this message when it cannot fulfill the message request made by the remote terminal:

- the first word is always set to 0,
- the 12 least significant bits of the 2nd word indicate the decimal value 1,
- the 4 most significant bits of the 2nd word indicate the cause of the refusal:
 - ♦ 1100 indicates that the controller does not know the request formulated,
 - ♦ 1101 indicates that a part of the other remote terminals concerned by the exchange are not available (switched off, failure etc.)
- the 3rd word contains the number of the request refused: this number is equal to the number set in the vector word.

When a remote terminal receives a refusal message, the associated transfer layer interrupts the latency time delay counter and informs the application. The connection must then be purged, i.e. everything that concerns the request at transfer level must be “forgotten”. (The purge concerns the data queue from the message to the request considered).

The application can then either abandon the request, or renew it according to the mechanism described in the paragraph entitled “Random exchange protocol (or exchange protocol on

request)” on page 43. The consequences of a point-to-point refusal must be studied case by case.

5.2.3.4.4.2 Generalized Refusal

The controller transmits 3 data words to sub-address 3 in point-to-point mode on the main line. This message is a random type message.

The controller transmits this message after an initialization/reset:

- unit switched on,
- return of power supply,
- reconfiguration.

It has then lost the list of random messages to be generated and sends, to all the remote terminals, a generalized refusal message (code: 1111). The third word transmitted does not have any particular meaning.

When the remote terminals receive the refusal message, their associated transfer layers interrupt the latency time delay counter and inform their applications. All the connections linked to all the requests must then be purged, i.e. everything that concerns the request at transfer level must be “forgotten”.

The sanction applied in this case (resetting all the exchange requests at the controller level to zero) is applied as a security measure. This is to avoid satisfying a request that has lapsed (in the event of a change in operational mode, for example). In the event that for a given remote terminal, the loss of exchange requests is dangerous, the data should be stored locally so the requests can be set again when the cancellation of the requests is indicated by the reception of the refusal message. The user level can set its request again if required.

Note: The general refusal message transmitted on the low-speed line after an initialization/reset of the controller enables it to indicate to all the remote terminals that none of their requests relating to either low-speed or high-speed exchanges can be taken into account.

5.2.3.4.5 Priority related to exchanges on request

These **priorities** occur at USER level. Five priority levels are defined for the entire system.

5.2.3.4.5.1 Times related to the sequencing of an exchange on request

For the transmission protocol for messages on request, two times are to be taken into account:

- The integration time for the request set by the USER level for an item of equipment. The integration time is the duration measured between the moment when this request is

submitted at the NETWORK level of an item of equipment and the moment when the queue is set up for this same request at the controller level. The value of this time depends on the integration mode of the request used. This time is known as **integration time**.

- The response time to this request. The response time is the time measured between the moment when the request is stored in a queue at the controller level and the moment when the corresponding message is received by its receiver(s). This time is known as **response time**.

5.2.3.4.5.2 Selection of priority level

Definition:

A maximum response time value is associated with each priority level Complying with transit time:

A transit time is specified at USER level for each message. The transit time is the time measured between the moment when this request is set at USER level for the requesting item of equipment and the reception of data at USER level of the receivers of the corresponding message.

As an initial approximation, it may be considered that the transit time is equal to:

$$\text{Transit time} = \text{Integration Time} + \text{Response time}$$

Two choices are possible when trying to comply with transit time:

- The first option is the integration mode of the request. This mode has a direct influence on the value of the integration time.
 - ◆ If this integration is carried out through periodic transmission by the controller of the polling message and only one request is on standby, the maximum integration time is equal to its transmission period.
 - ◆ If this integration is carried out at the request of the remote terminal by setting the Service Request bit, the maximum integration time is linked to the messages transmitted or received by the equipment. Insofar as the equipment receives or transmits a message at a given frequency, the maximum integration time is equal to the associated transmission period.
- The second option is the priority value associated with the message on request. This request is used to associate directly an objective response time sought which may possibly be exceeded by a few cycles in the event of overloading of the frame at a given moment. The five priority levels and the associated response times selected are:
 - ◆ **priority 0:** maximum objective response time sought = 3 msec
 - ◆ **priority H:** maximum objective response time sought = 20 msec

- ♦ **priority SH:** maximum objective response time sought = 80 msec (User Data Block with a size greater than the maximum size)
- ♦ **priority B:** maximum objective response time sought = 160 msec
- ♦ **priority SB:** any response time (User Data Block with a size greater than the maximum size)

In the first order, any request is integrated into the short cycle when it is received by the controller. Transit time = polling period.

In the second order, if the request has been postponed, the transit time is extended by the duration of a short cycle.

Selection of the transmission priority : A message transmission priority will be selected in agreement with DASSAULT AVIATION

5.2.3.4.6 Processing of messages transmitted on request

The processing of messages transmitted on request is characterized by the type of USER/NETWORK synchronization used as well as by the mode of production/consumption used.

5.2.3.4.6.1 USER/NETWORK Synchronization

The reception of any message transmitted on request is accompanied by a signal at USER level of the items of equipment concerned.

If an Exchange Request is refused, a signal is to be specified at the USER level of the item of equipment that has sent it (this item of equipment can be the transmitter or the receiver of the data message).

5.2.3.4.6.2 Production/Consumption of messages

- **Production of messages**

As regards data message transmitting equipment, two possibilities are feasible when this equipment also initiates the request:

- ♦ Flow type production mechanism. Each time a new value is produced from the same Network Data Block, the same Exchange Request is set. Storing the different values of the same Network Data Block requires that several data buffers be managed in a queue.
- ♦ Sampling type production mechanism. A single Exchange Request is set. Only the last value of the Network Data Block is transmitted. This production mechanism may be used to update the value of the Network Data Block between the time when the Exchange Request is carried out and the time when the data message is sent by the

equipment. In this case, a single data buffer is necessary. The new Network Data Block value deletes the previous one.

This description is applicable in the event that the requesting equipment is the transmitter of the data message. It is then necessary to make sure that the USER level provides the data first and then sets the corresponding Exchange Request.

In the event that the Exchange Request is at the initiative of the data message receiver, no synchronization can exist between the data production and the setting of the Exchange Request. **The production mechanism is of the sampling type.**

- **Consumption of messages**

To achieve consistency with the production of messages, the consumption mechanism is of the flow type if the production mechanism is of the flow type.

If the production mechanism is of the sampling type, the consumption mechanism can be either of the sampling type (this is the most common case), or of the flow type. The use of the consumption mechanism of the flow type enables the receiver in this case to avoid being controlled as far as the production speed of the transmitter is concerned.

5.3 Frames

5.3.1 Definition of the frame

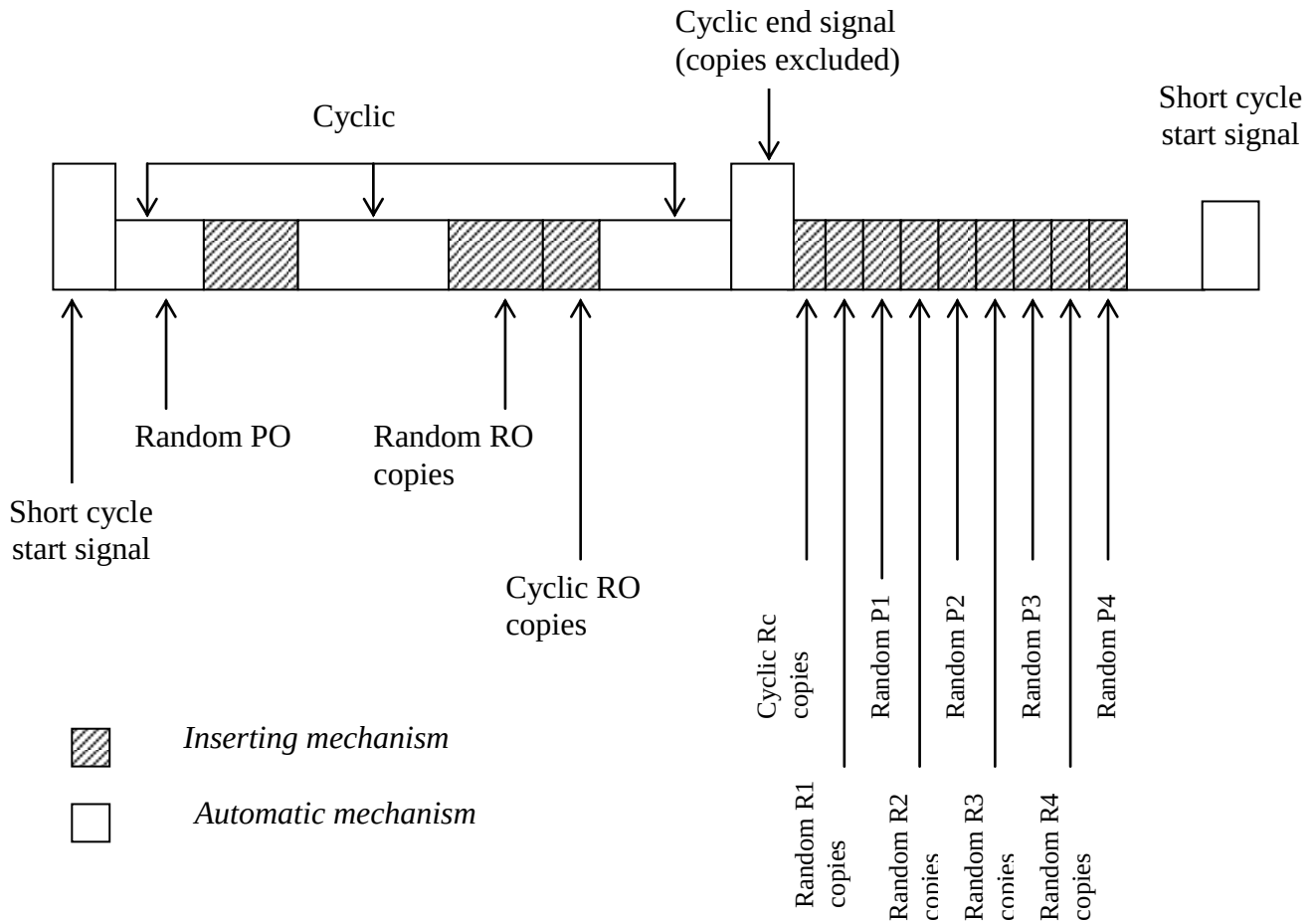
Frames are defined at BUS level. They correspond to the specification of the sequencing of bus messages.

5.3.1.1 Description

- Definition of the frame

8 short cycle tasks are defined. These tasks are made up of cyclic messages (50Hz, 25Hz, 12.5Hz, 6.25Hz) that are distributed equally in the 8 short cycles which constitute a long cycle. These tasks are generally of equal duration.

The following diagram illustrates a short cycle.



5.3.2 Sequencing of exchanges

The messages are **sequenced** on a bus out automatically and dynamically by the bus controller (with the assistance of the Exchange Sequencing Monitor).

Following this type of sequencing, the location of messages may differ from one software version to another, and the transmission period measured at the frame level is not stable from one cycle to another. The position in the frame may differ depending on whether or not the conditional cyclic messages or inserted. The transmission period of the message measured at frame level is thus different from the theoretical activation period of its transmission.

5.3.3 Rules for defining and executing frames

Each frame is made up of a set of lists. A list corresponds to a sequence of messages that possess the same transmission conditions, the same constraints and the same priority. The number of priorities and activation conditions is limited as much as possible.

Each one of these message sequences is activated on the bus according to its priority once its activation condition has been checked. A higher priority sequence may be inserted half way through a sequence with a lower priority (at the end of the transmission of the message in progress).

5.4 Identification and transfer of data

5.4.1 Equipment identification

Each piece of equipment on a given bus has two addresses:

- A physical equipment address, which is individual, for point-to-point exchanges
- A physical equipment address, common to all items of equipment and predefined by the MIL-STD-1553B (decimal address 31), for exchanges in broadcast mode.

Whatever the mode of data exchange (broadcast or point-on-point), the reception and transmission commands are similar. The distinction between the broadcasting of a message and its exchange in point-to-point mode is made by the identification carried out by the receivers.

- Either a single receiver is concerned and the receiver address is a physical equipment address,
- Or several receivers are concerned and the receiver address is the common physical equipment address. In the event that all the receivers are concerned, a **generalized** broadcast is performed. If this is not the case, the broadcast is **selective**. The TRANSFER layer recognizes the equipment involved.

Any equipment's terminal must be capable of decoding:

- Its physical equipment address (individual address) for the reception of a message sent in point-to-point mode.
- The common physical address (decimal value 31) for the reception of messages sent in broadcast mode.

All the remote terminals must be addressed explicitly. However, the controller terminal is addressed implicitly. There are therefore two types of addressing:

- **Implicit** addressing
- **Explicit** addressing

The controller terminals may thus be addressed implicitly on the bus to which they are connected via a controller terminal. This allows the same message to be sent to different receivers, one of these receivers being the Controller's host.

5.4.2 Data identification

5.4.2.1 Basic Identification

5.4.2.1.1 Presentation

This is the identification provided for in the MIL-STD-1553B protocol. The identification mode used is **physical** addressing. The Network Data Blocks are identified by a physical sub-address. In this case DLB=NDB. The identification of application data is thus equivalent to the identification of messages.

At the equipment level, a message is identified by its location via a physical address and the direction of the transmission (T/R bit).

- In T/R transmission = 1 32 sub-addresses are defined
- In T/R reception = 0

It is possible to receive messages in point-to-point mode (using the physical equipment address) or in broadcast mode (using the common physical address recognized by all the equipment).

- For first generation components, a single table is provided to receive messages in point-to-point or broadcast modes. If X messages are broadcast on the bus, only (32-X) possibilities remain for the reception of point-to-point exchanges.
- For second generation components, a table is associated with the physical address of each item of equipment and another table is associated with the common physical address. 32 messages can be sent in broadcast mode and 32 point-to-point exchanges can take place.

5.4.2.1.2 Upgrade capability

In order to facilitate any changes in the WDNS, no restrictions may be allowed regarding message transmission and reception capacity indicated in the MIL-STD-1553B standard. Any connected equipment must be capable of receiving and transmitting 32 maximum size messages.

5.4.2.2 Sub-address assignment rules

Only DASSAULT AVIATION can assign sub-addresses.

5.4.2.3 Sub-address assignment report

5.4.2.3.1 Sub-addresses with predefined use

5.4.2.3.1.1 MIL-STD-1553B predefinition

- Sub-addresses '0' and '31' identify a mode code and thus cannot be used to identify a Link Data Block.
- Sub-address '30' is used for equipment dialog tests.

5.4.2.3.1.2 System use predefinition

Note: This predefinition may be modified.

- Sub-address '1' is reserved for the high-speed line of a 3910 bus on the Rafale or for a store identification message (according to the MIL-STD-1760 standard) on a 1553B store bus.
- Sub-address '2' is reserved for sending the system time.
- Sub-address '3' and '4' are reserved for the other services provided by the transfer protocols.
 - ◆ Sub-address '3' is reserved for Class 2 refusal and acknowledgement messages.
 - ◆ Sub-address '4' is reserved for the downloading by the bus of the software application of an item of equipment.
- Sub-address '6' is reserved for the maintenance mode code for an item of equipment.

5.4.2.3.1.3 Provisions

The list of predefined sub-addresses can be added later (taking into account for example the broadcast of the maintenance mode word). To this effect, two supplementary sub-addresses, sub-addresses '5', are reserved as a provision.

5.4.2.3.2 Available sub-addresses

Sub-addresses '7' to '29' (23 different sub-addresses) are thus available for all the other data transfers on the bus for all connected equipment.

5.4.2.3.2.1 Messages associated with predefined sub-addresses

- **Polling message**

Use of the "Transmit vector word" mode code

CE(RT, Cm='10000') - ST(RT) D(vector)

The standard abbreviation of this message is **TVW(RT)** (Transmit Vector Word)

- **Refusal messages**

- ◆ **Refusal message in point-to-point mode**

CR(RT,SA=3,ND=1) D(refusal) - ST(RT)

- ◆ **Refusal message in broadcast mode**

CR(31,SA=3,ND=1) D(refusal) - ST(RT)

The definition of this message is found in the paragraph entitled "Logical addressing of Network Data Blocks (NDB)" on page 58.

Messages associated with synchronization/dating mechanisms

- ◆ Signals are broadcast using the "Synchronize with data word" mode code
CR(31,Cm='10001') D(signal)

The standard abbreviation of this message is **SYD(31)** (Synchronize with Data). The definition of the data word is found in the paragraph entitled "SYNCHRONIZATION / DATING" page 66.

- ◆ The time is sent by the following message:

CR(31,SA=2,ND=2) D (network time)

The standard abbreviation of this message is **CR(31,2,2)**. The definition of the two data words is found in the chapter entitled "SYNCHRONIZATION / DATING" page 66.

- Message relating to equipment to bus connection test

Use of the "Transmit Status Word" mode code

CE(RT,Cm='00010') - ST(RT)

The standard abbreviation of this message is **TSW(RT)** (Transmit Status Word)

- **Equipment dialog test messages**

- ◆ The computer sends a profile
CR(31,SA=30,ND=1) D (profile)

The standard abbreviation for this message is **CR(31,30,1)** This profile is encoded in a data word and expressed in hexadecimals: 5AA5, A55A or AAAA.

These values are provided for information purposes only and DASSAULT AVIATION reserves the right to modify them at any moment.

- ◆ Return of this profile by each item of equipment:
CE(RT,SA=30,ND=1) - ST (RT) D (profile)

The standard abbreviation for this message is **CE(RT,30,1)**

This list can be added to at a later time.

5.4.2.4 Complementary identification

5.4.2.4.1 Possible methods

5.4.2.4.1.1 Logical addressing of Network Data Blocks (NDB)

The **logical** addressing mode is not provided for in the MIL-STD-1553B protocol. It involves identifying not the location of the Network Data Blocks but the Network Data Blocks directly.

This addressing mode is used for the random exchange protocol to identify the exchange request (vector word coding). For transmission commands, the only addressing used is physical addressing. The correspondence table (logical addressing <-> physical addressing) is installed at the controller level. The priority of the exchange request is associated with the identification number of the request.

In order to simplify the decoding of the exchange requests, the identification numbers attributed are consecutive. 1024 exchange requests are provided for by the bus. This suggests a coding of 10 right-justified bits in the vector word (bits 6 to 15, imsb: bit 6, ilsb: bit 15), in compliance with the DGT document no. 81070 "General coding rules".

These exchange requests (which correspond to the connection reference) are either simple or multiple. The latter case corresponds to the implementation of a high-speed and low-speed segmentation mechanism. This mechanism initiates

- The multiple transmission requests: after a remote terminal request has been sent, the controller inserts the same random message several times into the frame.
- The requests for the transmission of acknowledgement messages used during segmentation is carried out with an explicit flow test which is used to reduce the size of the terminal memory necessary (class 2) in the event that large volumes of data are being transferred.

This complementary data is contained in the "TPDU-DT number" field in the vector word (bit 1 to 5, imsb: bit 1, ilsb: bit 5).

If the TPDU-DT number is equal to 0, this indicates an acknowledgement transmission request, and the connection reference number corresponds to the acknowledgement message identifier.

The value 31 for the TPDU-DT field is reserved.

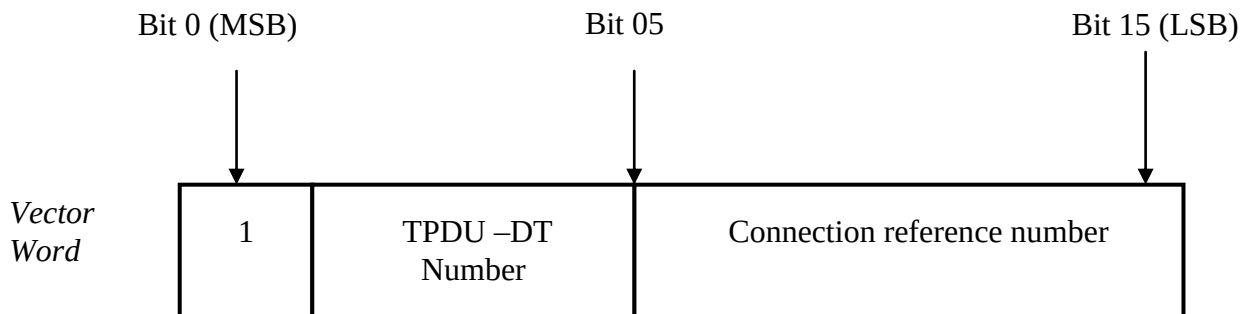
For any other value, the TPDU-DT field number corresponds to the multiplicity order of the request (which thus varies from 1 to 30).

The detailed description of the transfer layer segmentation mechanism is found in DGT document 97994, "Aircraft BUS 1553B and 3910, Additional Transfer Level Segmentation Mechanisms".

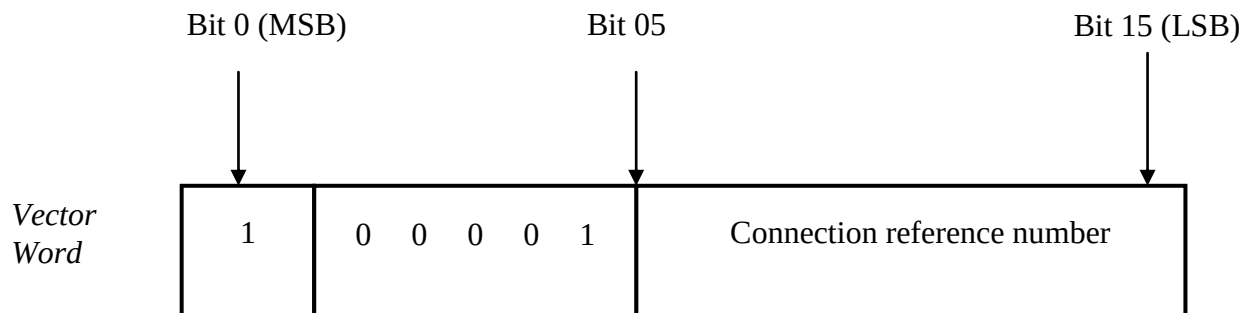
For reasons of compatibility with the appendices relating to data transmission problems on a MIL-STD-1553B bus when standardizing the aircraft store interface, the coding of the 0 bit is as follows:

- bit 0 = 0: the format of the vector word is the format standardized at MIL-STD-1760 standard level
- bit 0 = 1: the format of the vector word is the format that complies with the GAM-T-110 standard (ex GAM-T-103B) and corresponds to the ‘free user’ format proposed by the MIL-STD-1760 standard

In the second case, the vector word is encoded as follows:



Example: For any equipment setting non-1760 simple requests (multiplicity order of the request equal to 1), the vector word is encoded as follows:



These identification numbers are assigned exclusively by DASSAULT AVIATION.

Note: the use of logical addressing may be extended to the identification of an application task, or a subframe, etc. Task requests are thus usually set.

5.4.2.4.2 Extended physical address of the Network Data Blocks (NDB)

The number of sub-addresses remaining to identify the Network Data Blocks for a remote terminal is sufficient in the majority of cases. The distinct use of sub-addresses according to the type of command (transmission or reception) enables 23 distinct Network Data Blocks to be differentiated at the remote terminal during transmission (explicitly or in broadcast mode), 23 distinct Network Data Blocks in explicit reception or 23 Network Data Blocks in broadcast mode).

The bijective relation (sub-address <-> Network Data Block) limits the data volume that can be transmitted or received by an item of equipment. For this reason certain methods known as **extension** methods that increase volume are to be used.

Thus, certain items of equipment may receive or transmit a greater volume of data.

5.4.2.4.3 A brief summary of Addressing

A Network Data Block can be identified at bus level in two different ways:

- Through physical addressing
- Through logical addressing

Use of physical addressing:

A Network Data Block is transmitted by transmitting equipment and received by one or more items of receiving equipment. Depending on whether a centralized exchange (use of the implicit addressing for the controller computer) or a decentralized exchange is initiated, the Network Data Block is identified uniquely at bus level, either with respect to the transmitting equipment or the receiving equipment, or with respect to both.

1. Transmitting identification

- the transmitting address
- the direction of the transmission (transmission)
- the transmission sub-address
- the extension of the transmission sub-address (optional)

2. Receiving identification

- the receiving address
- the direction of the transmission (reception)
- the receiver sub-address
- the extension of the receiver sub-address (optional)

The bus controller only sets the mode codes. The sub-address extension is a local mechanism, processed by the NETWORK level of the equipment.

For real-time monitoring reasons, a Network Data Block must be able to be identified only by the data contained in the message (in the command phase and possibly in the data phases) irrespective of the type of equipment addressing used (explicit or implicit addressing).

Use of logical addressing:

The Network Data Block is identified solely at the bus level.

5.4.3 Sub-address extension rules

5.4.3.1 Sub-address multiplexing rules

If sub-address multiplexing is necessary for an item of equipment connected to the bus, the bus will be limited to indispensable sub-addresses only.

Two multiplexing possibilities are to be used:

- Multiplexing of a receiver sub-address
- Multiplexing of a transmission sub-address

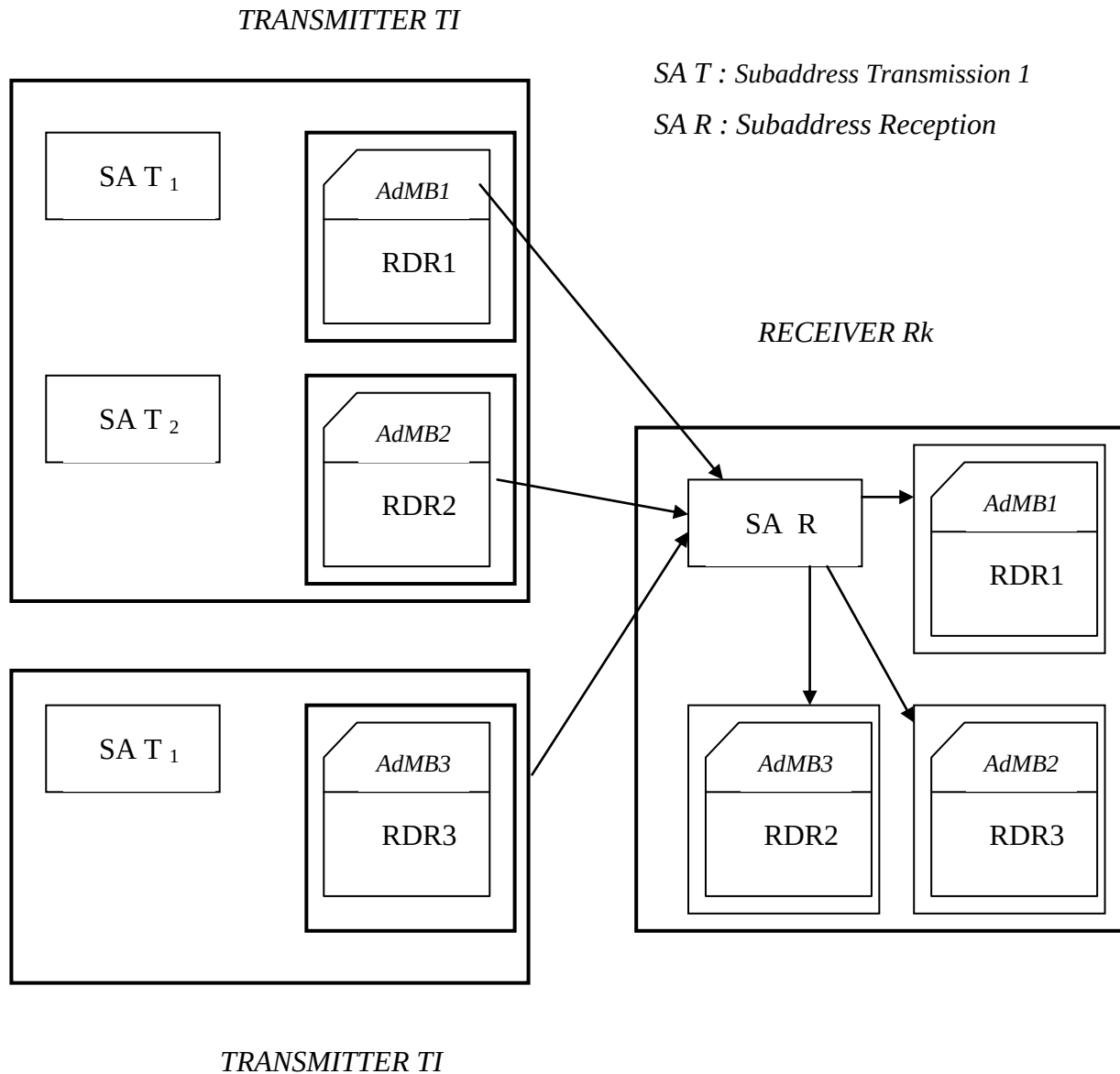
The need to implement a sub-address extension must be specified by DASSAULT AVIATION.

5.4.3.1.1 Receiving sub-address multiplexing

The implementation of this multiplexing requires the setting of a header. This header enables each receiver to identify the Network Data Block received.

No constraint is allowed on the sequencing of such messages. The receiving equipment must thus be capable of receiving on the same sub-address but for different sub-address extensions several consecutive messages without risk of data being deleted. This means that a buffer must be assigned to each data block.

The receiving sub-address multiplexing is represented in the following diagram:



Note 1: Sub-address reception multiplexing does not require any particular processing at NETWORK level for the transmitter. The header, for any message, is considered to be two data words, with predefined and constant values, set at USER level for the transmitting equipment.

Note 2: further to note 1, sub-address extension is not possible in broadcast mode. Insofar as the sub-address extension mechanism is optional, a misinterpretation of the data by a receiver that is not concerned may occur.

5.4.3.1.2 Multiplexing of a transmission sub-address

An item of equipment may use the same transmission sub-address to transmit different data blocks.

This multiplexing is only possible for messages transmitted on request. In automatic transmission, the controller has no means to identify the User Data Block to be exchanged and consequently the corresponding Network Data Block(s).

Two cases are to be taken into consideration for the transmission of this block which may initiate the transmission of several messages:

- transmission on the transmitter's request
- transmission on the receiver's request

5.4.3.1.2.1 Transmission on the transmitter's request

When transmission is carried out place on the request of the transmitter, the User Data Block to be transmitted is correctly identified and is supplied at Network level.

- In the event that decentralized exchanges are used (the receiver is not the controller), no complementary information to be set in the data phase is necessary to recognize the Link Data Block which is correctly identified using the reception command.
- In the event that centralized exchanges are used (the receiver is the controller), the command phase is made up of the transmission command word only, which is insufficient to identify the Network Data Block conveyed during the monitoring of the bus.

In order to make identical data phases available when the sub-address extension method is used, a header is systematically entered in the case of transmission sub-address multiplexing.

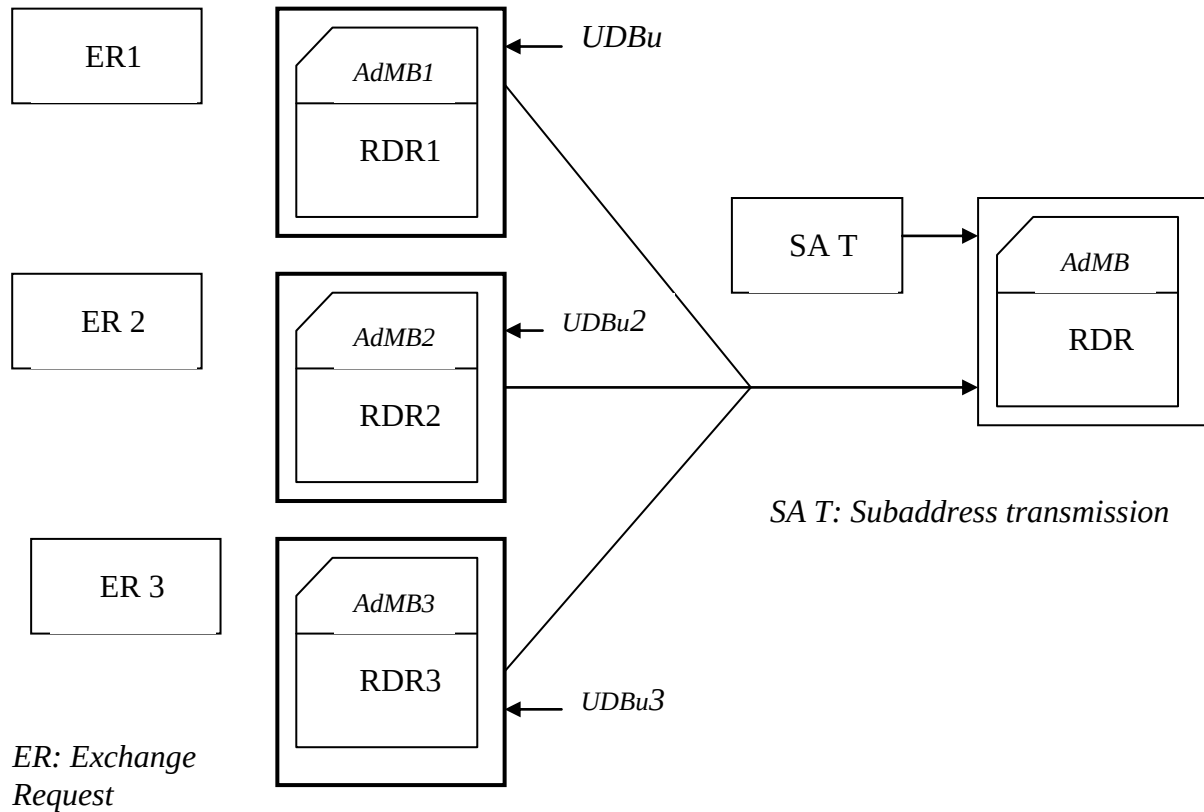
5.4.3.1.2.2 Transmission on the receiver's request

No transmission sub-address multiplexing can be performed because the controller cannot identify the Network Data blocks that correspond to the Application Data Block to be exchanged.

5.4.3.1.2.3 Conclusion

The transmission sub-address extension is limited to exchanges requested by the transmitter and requires the setting of a header.

The transmission sub-address multiplexing is represented in the following diagram:



Note: Sub-address multiplexing during transmission does not require any special processing of the header at the NETWORK level of the receiving equipment. The header is treated as two data words, which are not processed at the USER level of the receivers.

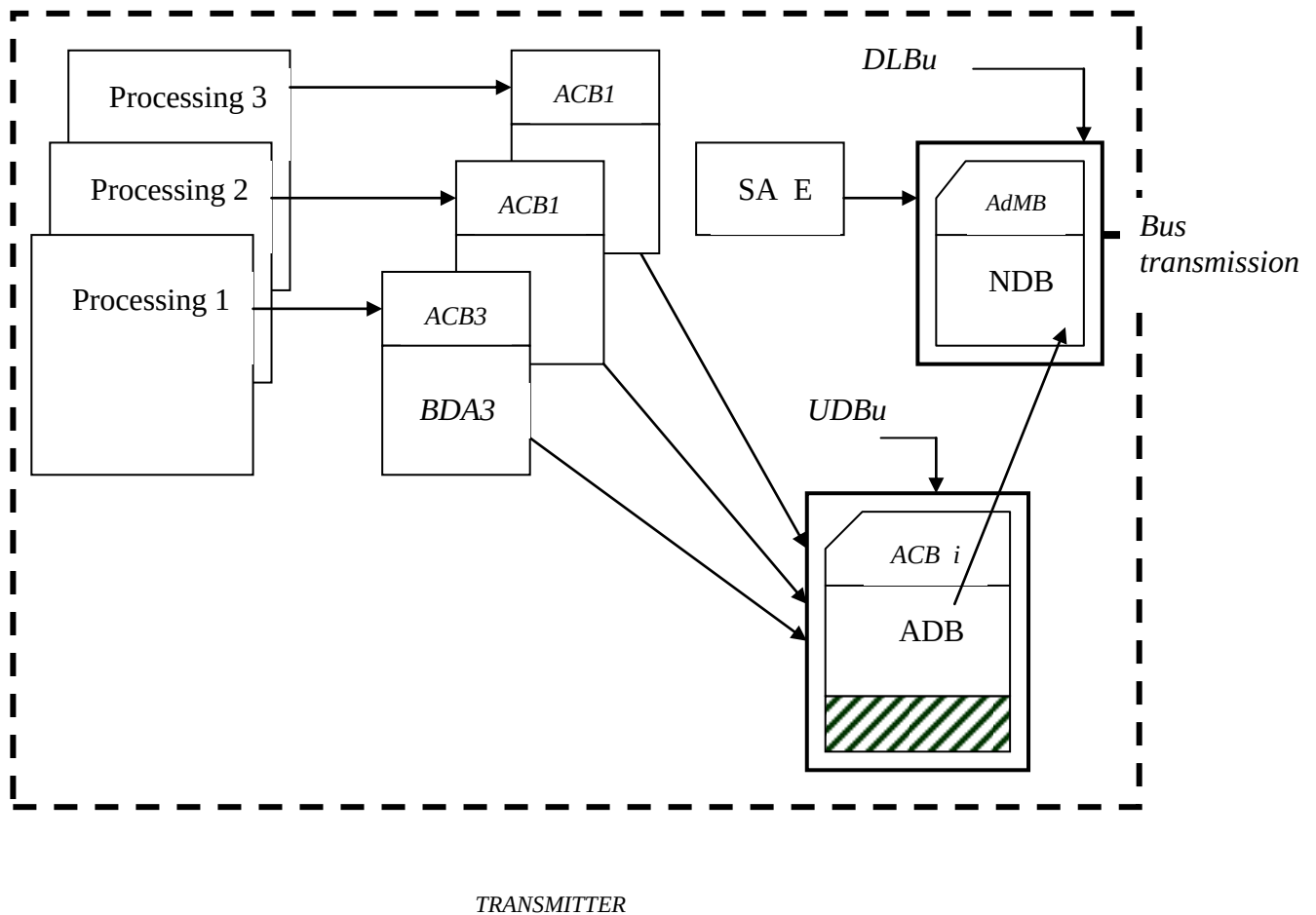
5.4.3.1.3 Multiplexing during transmission and reception

If the same header is used at the same time for transmission and reception sub-address extension, the simultaneous extension of sub-address transmission and reception requires the use of the same sub-address transmission and reception extension value.

5.4.4 Rules for using wagonnets

The wagonnet method involves time multiplexing at the “application” level of the User Data Blocks, in order to transfer on the bus, using the same message, only that which is required at a given moment.

Wagonnets are used when **only one among many** independent data blocks generated by an item of equipment can be transmitted on the bus, according to a determined operational condition (mode, configuration, etc.).



6. SYNCHRONIZATION / DATING

6.1 Definition of needs

Synchronization is used to sequence the processing operations executed in separate items of equipment.

Dating is used to measure the aging of data. Parameter dating may be carried out at several levels, this involves awareness of one of the following three dates:

- ◆ sensor data value sampling date
- ◆ processing end date
- ◆ date when output processing data is made available to the Bus Interface Unit for transmission.

6.2 Possible methods

6.2.1 Synchronization

Two methods are available:

- sending **signals**, which are specific or received simultaneously by all equipment
- sending **system time**

The term “system time” is the result of the fact that at the terminal level for each item of equipment a **local system time** is maintained. This local system time is controlled by a reference **master system time**.

6.2.2 Dating

Dating can be carried out explicitly or implicitly.

- **Explicit dating** is carried out using the system time.

At transmitter level, explicit dating is used to date the data transmitted (i.e. generation date), at receiving level it is used to date process initiation for example. The aging of the data is obtained by calculating the difference between these two dates.

- **Implicit dating** can be carried out as follows. At transmitter level, the generation of the data takes place at a known date (in relation to the signal). At receiver level, either the processing is performed at a known, predefined date (reference: signal or system time), or it is carried out at a non-predefined date (using system time).

6.3 Time and signal broadcasting mechanism

6.3.1 At bus level

The synchronization and system time signals are generated by the controller. This data is thus transmitted using messages sent alternately on each bus line.

6.3.1.1 Sending signals

6.3.1.1.1 Non-specific synchronization signals

On the main line, the controller transmits without guard time (i.e. as a priority) to all items of equipment the “Synchronize with data” mode code with a 10001 binary code. This message is cyclic and its frequency is 50Hz. The associated data contains the short cycle number within the long cycle (numbering from 0 to 7). The data is right-justified (use of bits 15 to 13, imsb: bit 13, ilsb: bit 15). The bits not used are forced to 0.

This synchronization signal is the first message of the new short cycle transmitted on the bus. When the time provided by the controller for each of the buses reaches a multiple value of 20msec, a new short cycle begins. However, the controller finishes transmitting the message in progress before transmitting the synchronization signal.

The principle of immediate retransmission for erroneous messages is not used. Thus, the maximum interval for a synchronization signal sent on the bus in relation to the theoretical signal at the start of the short cycle is equal to the longest message that can be transmitted, i.e. 800µsec for a 32-word message (approximately 1msec for the transmission of a 1024-word message on the Rafale 3910 rapid channel).

Upon reception of a synchronization signal, all the remote terminals are aware that all the messages which follow up to the next synchronization signal belong to the same short cycle.



6.3.1.1.2 Specific synchronization signals

The method for the transmission of specific synchronization signals to one or more items of equipment from the same bus involves the transmission by the bus controller of the message associated with the “synchronization” mode code (mode code = ‘00001’) transmitted explicitly and/or in broadcast mode.

These specific signals correspond to a remote interruption service.

6.3.1.2 Sending the system time

The master time bus controller transmits a message used to send the network time (the corresponding message can be positioned anywhere in the long cycle, from one cycle to another). This corresponds to the current value of the master system clock.

This message is transmitted on the main line in broadcast mode by the controller to all remote terminals with a decimal sub-address of 2. It is cyclic with a frequency of 6.25 Hz. The system time is binary-coded in 2 16-bit words and the LSB has a value of 10μs.

The system time is supplied by the time master in 32 bits (real data coded in 2 data words). This data is provided by the **master system clock**. No validity bit is associated with the time value without secure transmission.

The value of the **local system clock** corresponds to the value of the master system clock at the moment that transmission ends for the last bit of the message on the bus.

System time may enable items of equipment to date certain data as well as to trigger application level processing.

System time is controlled by a single master time which depends on the aircraft.

6.3.2 At system level

The computer controlling the principal 1553B buses of the WDNS is the master of the time it sends broadcasts via its controller terminals.

System time is supplied for synchronization and application dating purposes. As a result, a single reference time will be provided for equipment, even if this equipment is connected to several buses. The bus on which the system time is taken into account will thus be defined for each item of equipment.

6.4 Methods selected

6.4.1 Synchronization signals

For data coherence problems at system level, the start of each short cycle is explicitly marked by the transmission of the non-specific synchronization signal. This non-specific synchronization signal is used for the simultaneous activation of all the 50Hz processing operations in the different items of equipment.

The broadcasting method for specific signals is not used as a general mechanism for synchronization/dating problems.

6.4.2 Local system clock

A local system clock is provided at NETWORK level for each item of equipment. This clock is intended to carry out the dating of data, and the synchronization of processing operations if required.

The characteristics of this local clock must be identical to those of the master clock which is situated in the master time computer. (value coded in 32 bits, LSB value equal to 10 μ s). The updating mechanism of this clock in relation to the master system clock is the mechanism described in the paragraph entitled "Sending the system time" on page 68.

The difference between the values of the two local times provided in the two items of equipment connected to the network will be at most 100 μ s.

7. MONITORING OF THE BUS

7.1 Monitoring of exchanges

To guarantee correct transmission of data exchanges between items of equipment, different **monitoring techniques** are implemented simultaneously. Each one of them is involved with a particular aspect of the transmission.

These monitoring techniques are intended to detect any error than may occur during the transfer of data and any failure that may affect the transmission system.

They are divided into 2 groups:

- Checks of the transmission of data exchanged
- Checks of the bus coupling function for each item of equipment

The checks of the transmission of data exchanged are intended to detect **transmission errors**. The purpose of the bus coupling function checks is to detect **dialog failures**.

7.1.1 Checks of the transmission of data exchanged

Exchanges are continuously monitored via transmission checks performed on all the data exchanges between equipment.

Different checks are applied at LINK layer level and at TRANSFER layer level.

7.1.1.1 *Transmission checks carried out by the LINK layer*

At LINK layer level, these checks are performed by the bus controller and by the receiving remote terminals taking part in the exchange. They are described in the general rules (Doc DGT 95088, "Transmission monitoring"). The command words and data words sent over the bus are validate at the remote terminals, and messages are validated and status words analyzed at the bus controller.

7.1.1.2 *Transmission checks carried out by the TRANSFER layer*

At TRANSFER layer level, these checks are performed by the transmitter and by the receiver(s) only irrespective of the maintenance mode of the associated terminals. These involve checking duplication, sequencing and acknowledgement of data blocks and are described in the general rules (Doc DGT 95088 "Progress of a data transfer").

7.1.2 Checks of the bus coupling function

The bus coupling function checks validate the equipment's connection to the bus and the operation of the terminals when the system is in operational mode.

These checks are carried out in two different ways. The first involves carrying out tests via the bus by the bus controller. The second implements other tests performed internally for each terminal. Each type of test is carried out independently.

7.1.2.1 Check of the bus coupling function

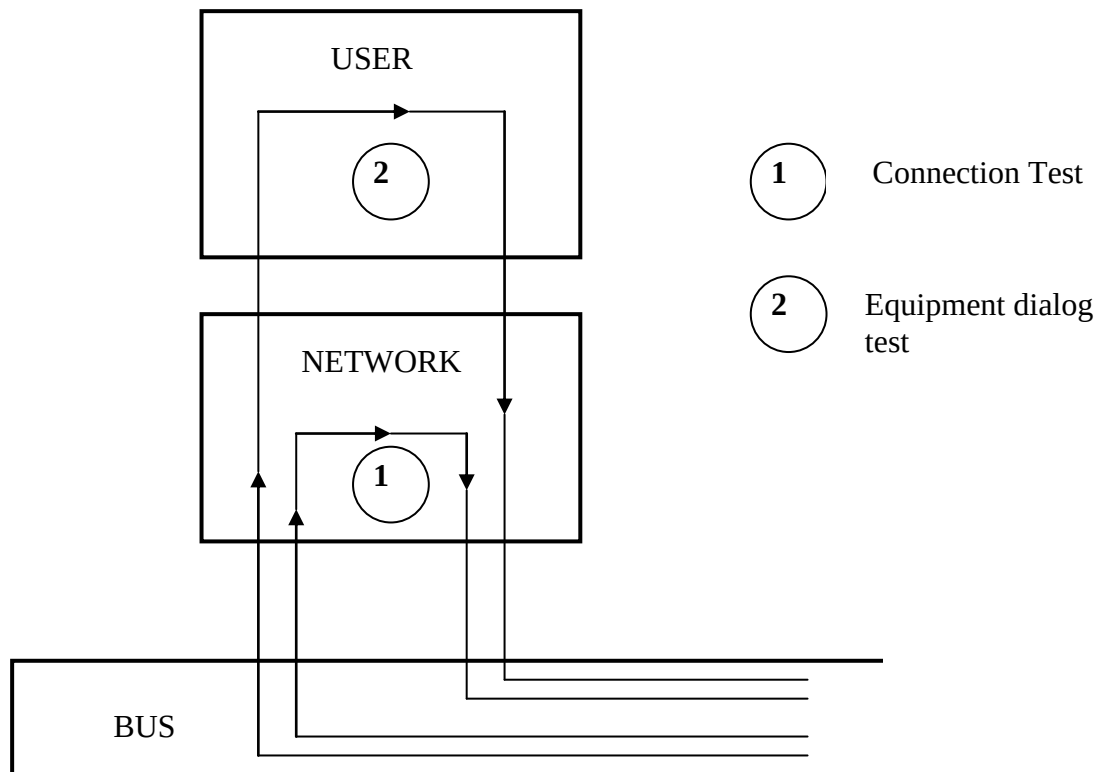
This paragraph is a brief overview of the general rules (Doc DGT 95088, "Remote terminal status acquisition").

The bus controller acquires the status of the remote terminals using specific non-interfering test procedures. These procedures are used to check the behavior of the terminals and more specifically the correct operation of the two following interfaces:

- The NETWORK/BUS interface using a connection test
- The NETWORK/USER interface using an equipment dialog test

Note: No local test of the network entity on the user entity is specified at system level; its implementation is left up to each manufacturer.

These two tests are known as **network tests** because they test the two interfaces that are concerned by the NETWORK level.



1. Connection test

This test checks the NETWORK/BUS interface which ensures the connection between the equipment and the bus. It is carried out by transmitting a message with the “transmit status word” mode code to the remote terminal to be tested.

Note: The “transmit last command ” mode code is not used to check the exchanges nor the network. But it may be used in special situations.

2. Equipment dialog test

This test checks the integrity of the connection between the BUS and the USER level, thus allowing the NETWORK/USER interface to be checked. It is carried out as follows:

3. The equipment dialog test comprises 2 messages:

- On the main line and in broadcast mode, the controller sends all the remote terminals, at sub-address 30, a 6.25Hz cyclic message. This message contains a data word which contains a profile with a value that is alternately A55A, 5AA5 and AAAA.

- On the main line and on the secondary line, each remote terminal sends, to the controller, at sub-address 30³;

A 6.25Hz cyclic message. This message contains the copy of the profile received from the controller 2 long cycles earlier.

- The transfer of this word by the equipment's application layer processor into a transmission memory position different from the receiving position;

Between the reception and the transmission of the profile, the host processor of the remote terminal must without fail transfer the profile from the receiving memory to the transmission memory in order to produce a read memory access and a write memory access.

The 2 messages which compose the test have no setting constraint in relation to each other. (The only constraint is the setting of the connection test: the equipment dialog test feedback must take place immediately after the connection test.). Thus during each long cycle⁴, the following always take place:

- The controller sends a new profile to the remote terminals
- Each remote terminal sends the controller the profile received 2 long cycles earlier.

This test allows the controller to check that the equipment connected to the bus it controls operates correctly. It is also used to select the line on which all the messages are sent (this known as the main line).

Note : For Rafale 3910, an identical test is carried out on a high-speed line. Thus, for each long cycle:

- The controller transmits in broadcast mode a 32-word block the first word of which contains a profile with a value that is successively 5AA5, A55A and AAAA;
- Each remote terminal transmits the profile received 2 long cycles earlier.

Note : Transmitting the profile received 2 long cycles earlier to the controller results in the fact that no item of equipment knows which profile to transmit following initialization/reset. No particular profile is imposed: the checks on equipment dialog test feedback begin for each item of equipment only from the 5th long cycle following each of its initializations/resets.

³To be compatible to a certain degree with paragraph 30.7 of Notice 2, the subaddress 30 must allow the transmission of a 32-word data block.

⁴Each long cycle begins with the mode code "Synchronize with data" for which the associated data contains the value 0.

7.1.2.2 Internal tests performed by the terminals

It is requested that the terminals possess a **self-test**. This self-test must be activated automatically when the terminal is switched on, and if possible continuously as a background task. It must also be activated on the request of the bus controller during reception of the “initialize self-test” mode code. In this case the remote terminal must indicate the start of its self-test to its associated item of equipment.

The result of the self-test must be indicated in the self-test word and in the status word. The self-test word must be defined in agreement with DASSAULT AVIATION. If an error is detected, the “terminal alarm” bit must be set to 1.

On reception of the “initialize the self-test” mode code the terminal must set the “busy” bit to 1 in its status word throughout the duration of the self-test and ignore any valid command addressing it other than the “transmit status word”, “disable transmitter” or “reset remote terminal” mode codes.

Note: The “initialize self-test” and “transmit self-test word” mode codes are not used for the operational checks of the exchanges and the network. However, they can be used in special situations (debugging and maintenance).

7.2 Use of monitoring

7.2.1 Errors taken into account at monitoring level

The following transmission errors may be detected by the bus controller during continuous monitoring of the transmission of the different messages at Link level, as well as during monitoring of the message used for the connection test:

- parity or symbol format error (Manchester coding)
- word format error
- no status
- addressing error
- no data
- number of incorrect data words
- response time > 14 μ s

The following status word errors may also be detected:

- busy bit set to 1
- terminal alarm bit set to 1
- equipment alarm bit set to 1

- one or more bits not set correctly

The connection test results take into account all of these errors. The results of the continuous monitoring on the messages at Link level takes into account only the transmission errors.

7.2.2 Determining equipment dialog statuses

The results alone of the monitoring carried out as coupling operation checks are taken into account to determine the dialog statuses of the remote terminals.

The bus controller determines, depending on the results of the 2 tests carried out (connection test and inter-equipment dialog test), the **dialog status** of each terminal connected to the bus. The dialog is declared to be in one of the following 3 states:

- In "**good working order**":
if the dialog with this remote terminal is possible on the 2 lines.
- "**minor dialog failure**" :
if the dialog with this remote terminal is only possible on one of the 2 lines.
- "**major dialog failure**":
if the dialog with this remote terminal is not possible on either of the 2 lines.

A minor dialog failure for a given line means that the transmitter and/or the receiver associated with this line have failed.

For each of these 2 tests, an error on a line is said to be confirmed as soon as at least one error has been detected on the same bus line twice in succession by the same test. Operation without error on a line is confirmed as soon as no error has been detected on the same bus line twice in succession by each one of the two tests.

7.2.3 Line redundancy

7.2.3.1 Justification

Line redundancy is necessary to inhibit equipment on a given line. It is also used to cover failures occurring at the physical level (transmitter/receiver failure, support cut-off, interference).

7.2.3.2 General description

Each of the buses is composed of two lines: an A line and a B line.

The **main line** is one of the two lines chosen to transmit all the messages in sequence, whether they are secure or not.

The **secondary line** is the other line on which the connection tests are sent.

Note: message redundancy is not used.

7.2.3.3 Line selection

A line is selected as follows:

- Main line = physical line A or B in flip-flop mode (a different main line is chosen after each powering up)
- **At the beginning of the long cycle** which follows the detection of the dialog failures during periodic monitoring carried out by the network tests.

7.2.3.4 Detailed description

7.2.3.4.1 Integration of network test results

The equipment dialog status is integrated at the beginning of each long cycle in order to find out which of the A or B lines has been chosen as the main line.

7.2.3.4.2 Integration of transmission monitoring results

The integration and processing of transmission errors detected during the transmission of messages is intended to cover transient errors, and not dialog failures. These failures are covered by the network tests.

This integration of transmission errors is not systematic. It can only be applied to certain messages with the consent of DASSAULT AVIATION.

In all cases, the line on which the message is transmitted for the first time is the main line, determined after the network test results have been received.

7.2.4 Disabling the equipment

Any equipment declared to be in “minor dialog failure” is disabled on the line where it does not function correctly.

Any equipment declared to be in “major dialog failure” is disabled on the main line.

Disabling is a sanction that follows different tests carried out by the bus controller.

This disabling on the line concerned must be terminated on reception of the random message which includes the “Disable transmitter” mode code (binary code 00100) transmitted by the bus controller **on the opposite line.**

Note 1: equipment disabled by the controller can only be enabled by the controller or during the next power-up (if the self-tests are correct)

Note 2: the equipment, regardless of its failure status, is enabled on the line(s) concerned during the network tests concerning it; or after another line has been selected.

8. MODE CODES USED

8.1 For controller terminals

The controller terminals must be capable of transmitting, in compliance with the 1553-B standard, all the following mode codes to any remote terminal, even if these mode codes have not been retained within the framework of this application:

Code	Function of the mode code	Type of transmission
'00000'	Dynamic bus control	*** Not used ***
'00001'	Synchronization	in point-to-point and broadcast mode
'00010'	Transmit status word	in point-to-point mode only
'00011'	Initialize self-test	in point-to-point and broadcast mode
'00100'	Disable transmitter	in point-to-point and broadcast mode
'00101'	Enable transmitter	in point-to-point and broadcast mode
'00110'	Disable terminal alarm bit	*** Not used ***
'00111'	Enable terminal alarm bit	*** Not used ***
'01000'	Reset remote terminal	in point-to-point and broadcast mode
'01001'	Transmit vector word	in point-to-point mode only
'01010'	Synchronize with data	in point-to-point and broadcast mode
'01011'	Transmit last command word	in point-to-point mode only
'01011'	Transmit self-test word	in point-to-point mode only
'10100'	Disable selected transmitter	*** Not used ***
'10101'	Enable selected transmitter	*** Not used ***

8.2 For remote terminals

Each of the remote terminals must at least implement and be able to respond to the following mode codes:

Code	Function of the mode code	Type of transmission
'00010'	Transmit status word	In point-to-point mode only
'00100'	Disable transmitter	In point-to-point and broadcast mode
'00101'	Enable transmitter	In point-to-point and broadcast mode
'01000'	Reset remote terminal	In point-to-point and broadcast mode
'10000'	Transmit vector word	In point-to-point mode only
'10001'	Synchronize with data	In broadcast mode

In addition, other mode codes must be implemented and processed as follows by the remote terminals with particular specificity:

- **'00011' – Initialize self-test:**

The mode codes must be implemented, used if necessary, if the remote terminal possesses self-tests (explicit reception or broadcast).

- **'00001' - Synchronize:**

The mode codes must be implemented in explicit reception, if the remote terminal requires particular synchronization signals (optional in broadcast mode).

- **'10010' – Transmit last command word:**

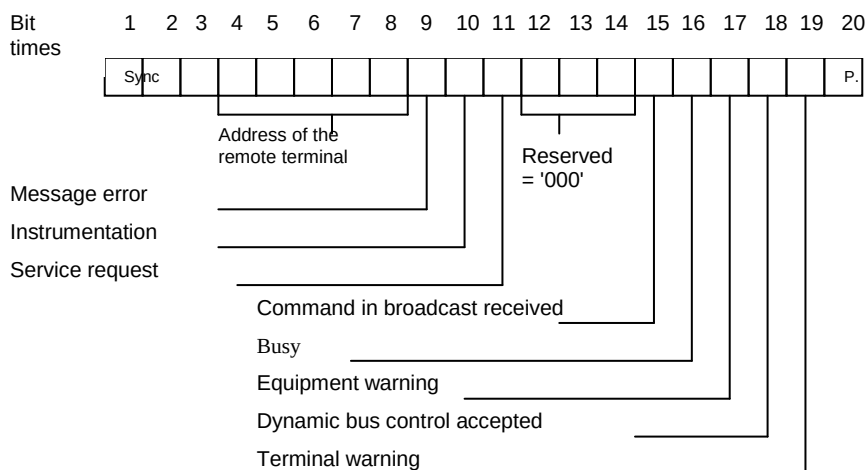
Optional implementation (in point-to-point mode only).

- **'10011' – Transmit self-test word:**

The mode codes must be implemented, used if necessary, if the remote terminal possesses self-tests (in point-to-point mode only).

9. EQUIPMENT STATUS WORD

9.1 Status word breakdown



9.2 Status word bits used

Each of the remote terminals must implement the equipment status word bits as follows:

- **Message errors:**

In compliance with the standard, this must always be implemented.

- **Instrumentation:**

Always set to 0.

- **Service request:**

Must always be implemented

- **Label command received:**

Must be implemented.

- **Busy:**

Must be implemented, must be set to 1, as long as the terminal is not available for the transmission or the reception of data, in particular when it is initialized or during an interfering self-test.

- **Equipment warning:**

Must be implemented, if the equipment containing the remote terminal possesses self-tests; must be set to 1, when the equipment has detected a failure in its operation through its self-tests, and remain at 1 as long as it is not re-declared to be in good working order. This bit does not take into account the functioning of the remote terminal.

- **Dynamic control of the accepted bus:**

Must remain at 0 in operational mode. It is intended to control this bit for the ground tool dialogs.

- **Terminal warning:**

Must be implemented, if the remote terminal possesses self-tests; must be set to 1, when the remote terminal is being initialized, or when it has detected through its self-tests a failure in its operation by remaining at 1 if it is not re-declared to be in good working order.

10. APPENDIX A: PROVISIONS FOR THE TRANSFER LAYER

A.1. IMPLEMENTATION OF THE TRANSFER LAYER

The services offered by the Transfer layer to each of the applications (users) communicating with each other via a 1553-B bus, as well as the dialog protocols necessary for the transfer of data between these applications, are standardized in the GAM-T-110 standard (ex GAM-T-103B).

This chapter presents the connections and the services selected for the implementation of the transfer layer.

A.2. CONNECTIONS SELECTED

The definition of and any modifications made to a WDNS must be checked with the contractor. The principal characteristic of the digital data transmission system is that it is a network fully predefined during the system development. Thus, for security reasons, it is necessary that all the data exchanges defined on the bus are known by the contractor. On each of the buses, all the data transfers between items of equipment are thus carried out on class 0 **open predefined** connections. The data is transferred:

- Either over **automatic** connections (transmission is activated by the network), in point-to-point or in generalized broadcast mode.
- Or over **generalized** connections (transmission is activated by the Users) in point-to-point or in generalized broadcast mode. Generalized broadcast only relates to centralized transmission connections.

The duplication and sequencing checks can be selected for the receiver on “automatic” connections, and the sequencing check on the “centralized transmission” connections.

Class	Operating mode	Transmission mode
0	automatic	point-to-point generalized broadcast
	centralized receiver	point-to-point
	centralized transmitter	point-to-point generalized broadcast

The choice of the type and of the checks selected for a specific connection is carried out at system level depending on the nature and the use of data exchanged.

A.3. SERVICES SELECTED

This paragraph describes the transfer layer services selected for all the WDNS equipment for the Rafale F1, F2 and MIRAGE 2000D. The GAM-T-110 (ex GAM-T-103) does not require implementation in the equipment of all the transfer layer services.

In addition, to facilitate the interconnection of similar items of equipment, it is not required to implement all the services selected for all the items of equipment. The choice, among the services selected, of those to be implemented for a given item of equipment must be made **in agreement with DASSAULT AVIATION**. This choice will be made in such a way that this item of equipment respects all the protocols defined and implemented at the Transfer layer level in compliance with the GAM-T-110 standard with regard to all the data transfers on the buses to which the item of equipment is connected.

Furthermore, the protocol used possesses centralized control. A greater number of services will be implemented for the controller terminal than for the remote terminals.

A.3.1. Communication services

A.3.1.1. Data transfer services

Only the data transfers “on connection” are selected.

These services enable a user to transmit data to or receive data from remote users on the connections selected above, when these are in open status.

These services require that the controller terminal implement resources for sending or receiving data over these connections.

A.3.1.2. Connection control services

Only suspension services (transfer connection becomes unstable) and restart services (transfer connection reopened) will be used.

These services enable a user to be informed of status changes in the predefined connections for which that user is an active member.

To facilitate these services, the controller terminal provides users of the same bus with the means to signal a status change of this connection to the active members of a predefined connection.

A.3.2. Synchronization services

A.3.2.1. Network time service

This service is indispensable for all equipment connected to the bus.

This service enables a user to acquire and to maintain a time reference

To facilitate this service, the controller terminal supplies the users of the same bus with the means to acquire a time reference common to these users.

A.3.2.2. Remote interruption service

This service is used to send specific synchronization signals on a bus to certain items of equipment connected to this bus. Taking into consideration the constraints for setting these signals in the frame, synchronization methods using the system time are preferable whenever possible.

This service enables a user to receive remote interruption data automatically through the bus, or to request the transmission of remote interruption data to the users of the same bus.

A.3.3. Supervision services

A.3.3.1. Configuration services

All the configuration services have been selected, except the download service which is not yet feasible.

These services enable a user to:

- charge the predefined parameters of his transfer entity locally, the parameters correspond to one or more operating configurations.
- reset his transfer entity on his request.
- request that his transfer entity be inserted into the network, in order to connect to the network.
- request that his transfer entity be removed from the network to disconnect it from the network while conserving the predefined parameters.

During the implementation of these services, the controller terminal will insert or remove the requesting user's transfer entity into or from the network.

A.3.3.2. Network observation services

These services enable a user to,

- request the acquisition of the network status and to receive the full status of the terminals on the network following his request.
- receive status changes for the terminals on the network.

To facilitate these services, the controller terminal allows users of the same bus:

- to generate and transmit on the user's request data pertaining to the full status of the terminals on the bus;

- to detect and indicate automatically to users on the bus, changes in the status of the terminals on the bus.

A.3.3.3. *Network checking services*

These services are only selected for the user associated with a terminal in controller mode.

These services enable the user to:

- request the reset of one or more remote transfer entities
- request the removal from the network of one or more remote transfer entities
- request the modification of the operational parameters of the network.

To carry out these services, the controller terminal offers means to

- order the reset of one or more network transfer entities.
- remove one or more transfer entities from the network.
- modify the operational parameters of the network.

The transfer entities on the bus must reset or withdraw themselves or modify their operational parameters on reception by the bus of the corresponding order.

A.3.3.4. *Maintenance services*

The local maintenance service is retained at the level of all the terminals. This service enables the user to check and validate the operational status of his terminal locally.

Distance maintenance service is only retained for the user associated with a terminal in controller mode.

This service enables the user to check and validate the operational status of remote terminals.

To facilitate this service, the controller terminal supplies its associated user with the means to check and validate the operational status of remote terminals.

A.3.3.5. *Purging service*

This service enables the user to invalidate the requests that it has previously provided for a given service.

A.3.4. *Summary of services retained*

The services retained at the equipment level, amongst all those offered in the GAM-T-110, have been divided into three categories depending on the terminal type contained in the equipment:

- Controller terminals
- Complex remote terminals (use of a specific processor associated with a terminal to assure communications functions)
- Simple remote terminals (no specific processor associated with the terminal)

This summary highlights:

- The services implemented (" * ")
- The services not retained (" - ")

The corresponding table is presented in the following pages :

Services	Eqpt with controller	Eqpt with complex remote terminal	Eqpt with simple remote terminal
COMMUNICATION			
* <i>Off-line transfer</i>	-	-	-
* <i>Predefined on-line transfer</i>			
0 – Automatic – Pt-to-pt	*	*	*
0 - Automatic - Broadcast	*	*	*
0 - Centr. Recap. – Pt-to-pt	*	*	*
0 - Centr. trans. – Pt-to-pt	*	*	*
0 - Centr. trans. - Broadcast	*	*	*
1 - Centr. trans. – Pt-to-pt	-	-	-
1 - Centr. trans. – Sel Brdcst	-	-	-
2 - Centr. trans. – Pt-to-pt	*	*	*
2 - Centr. trans. - Sel Brdcst	*	*	*
DYNAMIC	-	-	-
* <i>Connection control</i>			
Creation	-	-	-
Establishment	-	-	-
Opening	-	-	-
Closure	-	-	-
Restart	*	*	-
Suspension	*	*	-
SYNCHRONIZATION			
Network time	*	*	*

Remote interruption	*	-	-
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Services	Eqpt with controller	Eqpt with complex remote terminal	Eqpt with simple remote terminal
SUPERVISION			
* <i>Configuration</i>			
Loading	*	*	*
Downloading	*	*	-
Insertion	-	-	-
Removal	-	-	-
Reset	*	*	*
* <i>Network observation</i>			
Access to network status	*	*	-
Indication of status changes	*	*	-
Monitoring of equipment	*	-	-
* <i>Network check</i>			
Remote reset	*	-	-
Remote removal	*	-	-
Modification of parameters	*	-	-
* <i>Network Maintenance</i>			
Local maintenance	*	*	*
* <i>Purge</i>	*	*	*

A.4. Rules relating to the definition of parameters

Certain characteristics of the services provided to users by the TRANSFER layer are defined by the following parameters:

- **Service priority**

This parameter specifies the order of processing of services by the TRANSFER layer. This concept of priority associated with the service is independent of the concept defined for the random exchanges.

The priority of services selected is defined as follows from highest priority (1) to lowest:

1. Network time service remote interruption service

2. Purging service reset service
3. Loading service insertion service removal service restart service suspension service
4. On-line connection data transfer services
5. Network status access service status change indication service network check services local maintenance service distance maintenance service

- **Latency time delay**

This parameter specifies the maximum time tolerated by a user between the integration of the request and the supply of the corresponding confirmation for the service requested. It must be specified for the following services:

- On-line connection transfer for:
 - ◆ the receiver on the connections Centralized 0 class point-to-point receiver
 - ◆ the transmitter on the connections Centralized point-to-point transmitter or generalized Class 0 broadcast mode transmitter
- Remote interruption on request
- Loading
- Insertion
- Removal
- Reset
- Access to network status
- Network check
- Local maintenance
- Remote maintenance
- Purge

The latency time delay values are defined by par DASSAULT AVIATION.

- **Integrity Condition**

In order to be verified the integrity condition of a connection requires that all items of equipment “indispensable” to the connection are in operational status and not declared to be in “major dialog failure” by the monitoring tests on the network. In addition, if certain of these indispensable items of equipment are in “minor dialog failure” they must be on the same bus. The “indispensable” items of equipment in a transfer are:

- in point-to-point mode, the transmitter and the receiver.
- in generalized broadcast mode, the transmitter.

- in selective broadcast mode, the transmitter and all the receivers.

11. APPENDIX B: SYSTEM MESSAGES SENT OVER THE 3 BUSES

- System Time
- Equipment Dialog Test
- Synchronization Signals
- Request to Transmit Vector Word
- Request to Transmit Status Word
- Refusal Messages (point-to-point and broadcast modes)
- Network Status
- Remote terminal disable command
- Remote terminal enable command